

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

3.0 TDV6 ENGINE CONTROL MODULE
(ECM) - P0300-00 TO P06FB-96

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed via the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

The table below lists all diagnostic trouble codes (DTCs) that could be logged in the rear integrated control panel, for additional diagnosis and testing information refer to the relevant diagnosis and testing section. For additional information, refer to: [Electronic Engine Controls](#) (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0300-00	Random/Multiple Cylinder Misfire Detected - No sub type information	▪ Loss of compression ▪ Fuel leakage at fuel injector ▪ Other fuel injector related DTCs ▪ Fuel injector mechanical integrity ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Fuel injector failure	▪ Carry out a cylinder compress ▪ Check fuel injector for leakage ▪ Check engine control module injection related DTCs and ref relevant DTC index ▪ Check fuel injector for mechan integrity ▪ Refer to the electrical circuit d and check the fuel injector cir short circuit to ground, short c power, open circuit, high resis ▪ Inspect connectors for signs o ingress, and pins for damage : corrosion ▪ Check and install a new fuel in required ▪ Using the manufacturer ap diagnostic system carry o 'Injector replacement' rou ▪ Using the manufacturer appro diagnostic system reprogram injector codes ▪ Using the manufacturer appro diagnostic system clear all sto using the Diagnosis Menu tab retest
P0301-00	Cylinder 1 Misfire Detected - No sub type information	▪ Loss of compression ▪ Fuel leakage at fuel injector ▪ Other fuel injector related DTCs ▪ Fuel injector mechanical integrity ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Carry out a cylinder compress ▪ Check fuel injector for leakage ▪ Check engine control module injection related DTCs and ref relevant DTC index ▪ Check fuel injector for mechan integrity ▪ Refer to the electrical circuit d

		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Fuel injector failure	and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new fuel injector if required ▪ Using the manufacturer approved diagnostic system carry out the 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0302-00	Cylinder 2 Misfire Detected - No sub type information	▪ Loss of compression ▪ Fuel leakage at fuel injector ▪ Other fuel injector related DTCs ▪ Fuel injector mechanical integrity ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Fuel injector failure	▪ Carry out a cylinder compression test ▪ Check fuel injector for leakage ▪ Check engine control module for injection related DTCs and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Refer to the electrical circuit diagram and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new fuel injector if required ▪ Using the manufacturer approved diagnostic system carry out the 'Injector replacement' routine ▪ Using the manufacturer approved diagnostic system reprogram the injector codes ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

P0303-00	Cylinder 3 Misfire Detected - No sub type information	▪ Loss of compression ▪ Fuel leakage at fuel injector ▪ Other fuel injector related DTCs ▪ Fuel injector mechanical integrity ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Fuel injector failure	▪ Carry out a cylinder compress ▪ Check fuel injector for leakage ▪ Check engine control module injection related DTCs and ref relevant DTC index ▪ Check fuel injector for mechan integrity ▪ Refer to the electrical circuit d and check the fuel injector circ short circuit to ground, short c power, open circuit, high resis ▪ Inspect connectors for signs o ingress, and pins for damage a corrosion ▪ Check and install a new fuel in required ▪ Using the manufacturer ap diagnostic system carry o 'Injector replacement' rou ▪ Using the manufacturer appro diagnostic system reprogram i injector codes ▪ Using the manufacturer appro diagnostic system clear all sto using the Diagnosis Menu tab retest
P0304-00	Cylinder 4 Misfire Detected - No sub type information	▪ Loss of compression ▪ Fuel leakage at fuel injector ▪ Other fuel injector related DTCs ▪ Fuel injector mechanical integrity ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Fuel injector failure	▪ Carry out a cylinder compress ▪ Check fuel injector for leakage ▪ Check engine control module injection related DTCs and ref relevant DTC index ▪ Check fuel injector for mechan integrity ▪ Refer to the electrical circuit d and check the fuel injector circ short circuit to ground, short c power, open circuit, high resis ▪ Inspect connectors for signs o ingress, and pins for damage a corrosion ▪ Check and install a new fuel in required ▪ Using the manufacturer ap diagnostic system carry o

			'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram injector codes - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0305-00	Cylinder 5 Misfire Detected - No sub type information	- Loss of compression - Fuel leakage at fuel injector - Other fuel injector related DTCs - Fuel injector mechanical integrity - Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Fuel injector failure	- Carry out a cylinder compression test - Check fuel injector for leakage - Check engine control module injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Refer to the electrical circuit diagram and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and corrosion - Check and install a new fuel injector if required - Using the manufacturer approved diagnostic system carry out 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram injector codes - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0306-00	Cylinder 6 Misfire Detected - No sub type information	- Loss of compression - Fuel leakage at fuel injector - Other fuel injector related DTCs - Fuel injector mechanical integrity - Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Carry out a cylinder compression test - Check fuel injector for leakage - Check engine control module injection related DTCs and refer to relevant DTC index - Check fuel injector for mechanical integrity - Refer to the electrical circuit diagram

		- Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Fuel injector failure	and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new fuel injector as required - Using the manufacturer approved diagnostic system carry out the 'Injector replacement' routine - Using the manufacturer approved diagnostic system reprogram the injector codes - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0313-00	Misfire Detected With Low Fuel - No sub type information	- Low level fuel condition enabling run dry strategy - Other fuel related DTCs - Fuel leakage - Fuel level sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel level sensor failure	- Check the fuel level, add fuel as required - Check engine control module related DTCs and refer to relevant index - Check fuel system for mechanical integrity - Refer to the electrical circuit diagram and check the fuel level sensor for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new fuel level sensor as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0315-32	Crankshaft Position System Variation Not Learned - Signal low time < minimum	- The engine control module detected the low pulse is too narrow with respect to time - Crankshaft position sensor incorrectly installed - Crankshaft position sensor air	- Check crankshaft position sensor correct installation - Check crankshaft position sensor gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor

		gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ▪ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Crankshaft position sensor failure	▪ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new crankshaft position sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0335-00	Crankshaft Position Sensor "A" Circuit - No sub type information	▪ Crankshaft position sensor incorrectly installed ▪ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ▪ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Crankshaft position sensor failure	▪ Check crankshaft position sensor correct installation ▪ Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor ▪ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new crankshaft position sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0335-31	Crankshaft Position Sensor "A" Circuit - No signal	▪ The engine control module does not detect a signal which ought to be present ▪ Crankshaft position sensor incorrectly installed ▪ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ▪ Crankshaft position sensor	▪ Check crankshaft position sensor correct installation ▪ Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor ▪ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Crankshaft position sensor failure	resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check and install a new crankshaft position sensor as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0335-39	Crankshaft Position Sensor "A" Circuit - Incorrect has too few pulses	- Crankshaft position sensor incorrectly installed - Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor - Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Crankshaft position sensor failure	- Check crankshaft position sensor correct installation - Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor - Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check and install a new crankshaft position sensor as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0335-3A	Crankshaft Position Sensor "A" Circuit - Incorrect has too many pulses	- Crankshaft position sensor incorrectly installed - Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor - Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check crankshaft position sensor correct installation - Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor - Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion

		■ Crankshaft position sensor failure	■ Check and install a new crankshaft position sensor as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0335-64	Crankshaft Position Sensor "A" Circuit - Signal plausibility failure	■ Crankshaft position sensor incorrectly installed ■ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ■ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Crankshaft position sensor failure	■ Check crankshaft position sensor correct installation ■ Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, damaged teeth on rotor ■ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ■ Check and install a new crankshaft position sensor as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0336-31	Crankshaft Position Sensor "A" Circuit Range/Performance - No signal	■ The engine control module does not detect a signal which ought to be present ■ Crankshaft position sensor incorrectly installed ■ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ■ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Crankshaft position sensor failure	■ Check crankshaft position sensor correct installation ■ Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, damaged teeth on rotor ■ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ■ Check and install a new crankshaft position sensor as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

			using the Diagnosis Menu tab retest
P0336-64	Crankshaft Position Sensor "A" Circuit Range/Performance - Signal plausibility failure	■ Crankshaft position sensor incorrectly installed ■ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ■ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Crankshaft position sensor failure	■ Check crankshaft position sensor correct installation ■ Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, damaged teeth on rotor ■ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ■ Check and install a new crankshaft position sensor as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0339-2F	Crankshaft Position Sensor "A" Circuit Intermittent - Signal erratic	■ Crankshaft position sensor incorrectly installed ■ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ■ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Crankshaft position sensor failure	■ Check crankshaft position sensor correct installation ■ Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, damaged teeth on rotor ■ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ■ Check and install a new crankshaft position sensor as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

P0339-31	Crankshaft Position Sensor "A" Circuit Intermittent - No signal	▪ The engine control module does not detect a signal which ought to be present ▪ Crankshaft position sensor incorrectly installed ▪ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ▪ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Crankshaft position sensor failure	▪ Check crankshaft position sensor correct installation ▪ Check crankshaft position sensor gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor ▪ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new crankshaft position sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0340-31	Camshaft Position Sensor "A" Circuit Bank 1 or Single Sensor - No signal	▪ The engine control module does not detect a signal which ought to be present ▪ Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Camshaft position sensor or reference target positioning incorrect ▪ Camshaft position sensor failure	▪ Refer to the electrical circuit diagram and check the camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check the sensor and camshaft for damage, contamination, and mounting. Check camshaft timing specification ▪ Check and install a new camshaft position sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0341-64	Camshaft Position Sensor "A" Circuit Range/Performance Bank 1 or single sensor - Signal plausibility failure	▪ Camshaft position sensor incorrectly installed ▪ Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check camshaft position sensor correct installation ▪ Refer to the electrical circuit diagram and check the camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		- Connector is disconnected, connector pin is backed out, connector pin corrosion - Camshaft position sensor or reference target positioning incorrect - Camshaft position sensor failure	resistance - Inspect connectors for signs of ingress, and pins for damage or corrosion - Check the sensor and camshaft for damage, contamination, and mounting. Check camshaft timing specification - Check and install a new camshaft position sensor as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0342-31	Camshaft Position Sensor "A" Circuit Low Bank 1 or single sensor - No signal	- The engine control module does not detect a signal which ought to be present - Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Camshaft position sensor or reference target positioning incorrect - Camshaft position sensor failure	- Refer to the electrical circuit diagram and check the camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage or corrosion - Check the sensor and camshaft for damage, contamination, and mounting. Check camshaft timing specification - Check and install a new camshaft position sensor as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P034A-00	Crankshaft Position Sensor - Crankshaft Start Position Incorrect - No sub type information	- Crankshaft position sensor incorrectly installed - Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor - Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out,	- Check crankshaft position sensor correct installation - Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor - Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of

		connector pin corrosion ■ Crankshaft position sensor failure	ingress, and pins for damage and corrosion ■ Check and install a new crankshaft position sensor as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P034A-64	Crankshaft Position Sensor - Crankshaft Start Position Incorrect - Signal plausibility failure	■ The engine control module detected plausibility failures ■ Crankshaft position sensor incorrectly installed ■ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ■ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Crankshaft position sensor failure	■ Check crankshaft position sensor correct installation ■ Check crankshaft position sensor gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor ■ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, resistance ■ Inspect connectors for signs of ingress, and pins for damage and corrosion ■ Check and install a new crankshaft position sensor as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P034A-67	Crankshaft Position Sensor - Crankshaft Start Position Incorrect - Signal incorrect after event	■ The engine control module did not see the correct change of a parameter or group of parameters in response to a particular event ■ Crankshaft position sensor incorrectly installed ■ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ■ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected,	■ Check crankshaft position sensor correct installation ■ Check crankshaft position sensor gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor ■ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, resistance ■ Inspect connectors for signs of ingress, and pins for damage and corrosion ■ Check and install a new crankshaft position sensor as required

		connector pin is backed out, connector pin corrosion ■ Crankshaft position sensor failure	■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P034A-86	Crankshaft Position Sensor - Crankshaft Start Position Incorrect - Signal invalid	■ The engine control module has determined failures where some circuit quantity, reported via serial data, is not plausible given the operating conditions ■ Crankshaft position sensor incorrectly installed ■ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ■ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Crankshaft position sensor failure	■ Check crankshaft position sensor correct installation ■ Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor ■ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ■ Check and install a new crankshaft position sensor as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P034B-00	Crankshaft Position Sensor - Crankshaft Direction Incorrect - No sub type information	■ Crankshaft position sensor incorrectly installed ■ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ■ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Crankshaft position sensor failure	■ Check crankshaft position sensor correct installation ■ Check crankshaft position sensor air gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor ■ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ■ Check and install a new crankshaft position sensor as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

P034B-76	Crankshaft Position Sensor - Crankshaft Direction Incorrect - Wrong mounting position	▪ The engine control module has detected incorrectly installed components ▪ Crankshaft position sensor incorrectly installed ▪ Crankshaft position sensor air gap to target rotor excessive, debris on sensor face, damaged teeth on rotor ▪ Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Crankshaft position sensor failure	▪ Check crankshaft position sensor correct installation ▪ Check crankshaft position sensor gap to target rotor is correct, any debris from sensor face, or damaged teeth on rotor ▪ Refer to the electrical circuit diagram and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new crankshaft position sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0383-11	Glow Plug Control Module 1 Control Circuit Low - Circuit short to ground	NOTE: Glow plug control module ST pin connected to ground ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Glow plug control module circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagram and check the glow plug control module circuit for short circuit to ground ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0384-12	Glow Plug Control Module 1 Control Circuit High - Circuit		▪ Refer to the electrical circuit diagram and check the glow plug control module circuit for short circuit

	short to battery	NOTE: Glow plug control module ST pin connected to power ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Glow plug control module circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug control module failure	▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0401-00	EGR "A" Flow Insufficient Detected - No sub type information	▪ Other EGR related DTCs ▪ Exhaust gas recirculation valve - High pressure ▪ EGR cooler / system blocked ▪ Exhaust gas recirculation valve - High pressure ▪ EGR valve mechanical integrity ▪ Exhaust gas recirculation valve - High pressure ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check engine control module related DTCs and refer to relevant index ▪ Exhaust gas recirculation valve - High pressure ▪ Check EGR cooler / system for blockage ▪ Exhaust gas recirculation valve - High pressure ▪ Check EGR valve for mechanical integrity ▪ Exhaust gas recirculation valve - High pressure ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Using the manufacturer approved diagnostic system, perform road test. If the Exhaust gas recirculation valve DTCs and retest ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

P0402-00	EGR "A" Flow Excessive Detected - No sub type information	- Other EGR related DTCs - Exhaust gas recirculation valve - High pressure - EGR valve exhaust gas leakage into air intake system - Exhaust gas recirculation valve - High pressure - EGR cooler / system blocked - Exhaust gas recirculation valve - High pressure - EGR valve mechanical integrity - Exhaust gas recirculation valve - High pressure - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module related DTCs and refer to rele index - Exhaust gas recirculation valve - High pressure - Check EGR valve for exhaust gas leakage into air intake system - Exhaust gas recirculation valve - High pressure - Check EGR cooler / system for blockage - Exhaust gas recirculation valve - High pressure - Check EGR valve for mechanical integrity - Exhaust gas recirculation valve - High pressure - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. If the DTC returns, use the Inline diagnostic unit 2 diagnostic tool to clear the Exhaust gas recirculation valve DTCs and retest - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0403-13	EGR "A" Control Circuit/Open - Circuit open	- The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - EGR valve circuit open circuit, high resistance - Exhaust gas recirculation valve - High pressure - Connector is disconnected, connector pin is backed out, connector pin corrosion - EGR valve failure - Exhaust gas recirculation valve - High pressure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x0374 EGR Valve Position - Volts 0x03FB EGR Bank 1 - Commanded - % 0xF42C Commanded EGR Valve - % - Refer to the electrical circuit diagram and check the EGR valve circuit for open circuit, high resistance - Exhaust gas recirculation valve - High pressure - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved

			diagnostic system, perform ro Inline diagnostic unit 2 diagn Exhaust gas recirculation valve the DTCs and retest - Check and install a new EGR v required - Exhaust gas recirculation v High pressure - Using the manufacturer ap diagnostic system carry o path set-up routine' routin (0x4030/0x4051_01/0x40 0xF405) - Using the manufacturer ap diagnostic system carry o control module low pressu adaption clear' routine (0x - Using the manufacturer appro diagnostic system clear all sto using the Diagnosis Menu tab retest
P0404-77	EGR "A" Control Circuit Range/Performance - Commanded position not reachable	- The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment - EGR valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Exhaust gas recirculation valve - High pressure - Connector is disconnected, connector pin is backed out, connector pin corrosion - EGR valve failure - Exhaust gas recirculation valve - High pressure	- Diagnosis of this DTC may rec the manufacturer approved di system check datalogger sign 0x0374 EGR Valve Positio Voltage - Volts 0x03FB EGR Bank 1 - Con - % 0xF42C Commanded EGF - Refer to the electrical circuit d and check the EGR valve circu short circuit to ground, short c power, open circuit, high resis - Exhaust gas recirculation v High pressure - Inspect connectors for signs o ingress, and pins for damage corrosion - Using the manufacturer appro diagnostic system, perform ro Inline diagnostic unit 2 diagn Exhaust gas recirculation valve the DTCs and retest - Check and install a new EGR v required - Exhaust gas recirculation v

			High pressure - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure adaption clear' routine (0x4030/0x4051_01/0x4051_0xF405) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0405-16	EGR Sensor "A" Circuit Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - EGR valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Exhaust gas recirculation valve - High pressure - Connector is disconnected, connector pin is backed out, connector pin corrosion - EGR valve failure - Exhaust gas recirculation valve - High pressure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x0374 EGR Valve Position Voltage - Volts 0x03FB EGR Bank 1 - Commanded - % 0xF42C Commanded EGR Valve - % - Refer to the electrical circuit diagram and check the EGR valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Exhaust gas recirculation valve - High pressure - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve test the DTCs and retest - Check and install a new EGR valve if required - Exhaust gas recirculation valve - High pressure - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_0xF405) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

			diagnostic system carry out 'control module low pressure adaption clear' routine (0x00000000) Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0406-17	EGR Sensor "A" Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - EGR valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Exhaust gas recirculation valve - High pressure - Connector is disconnected, connector pin is backed out, connector pin corrosion - EGR valve failure - Exhaust gas recirculation valve - High pressure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x0374 EGR Valve Position Voltage - Volts 0x03FB EGR Bank 1 - Commanded - % 0xF42C Commanded EGR Valve Position - % - Refer to the electrical circuit diagram and check the EGR valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Exhaust gas recirculation valve - High pressure - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnose Exhaust gas recirculation valve and clear the DTCs and retest - Check and install a new EGR valve if required - Exhaust gas recirculation valve - High pressure - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051/0x4051/0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure adaption clear' routine (0x00000000) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

P0407-00	EGR Sensor "B" Circuit Low - No sub type information	▪ EGR valve circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ EGR valve failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals ▪ 0x0374 EGR Valve Position - Volts ▪ 0x03FB EGR Bank 1 - Commanded - % ▪ 0xF42C Commanded EGR Valve Position - % ▪ Refer to the electrical circuit diagram and check the EGR valve circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnose Exhaust gas recirculation valve operation. Clear the DTCs and retest ▪ Check and install a new EGR valve if required ▪ Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405) ▪ Using the manufacturer approved diagnostic system carry out 'control module low pressure EGR valve adaption clear' routine (0x4051_03/0x4051_04/0xF405) ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0407-16	EGR Sensor "B" Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Low pressure EGR valve circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out,	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals ▪ 0x05CD Low Pressure EGR Valve Position - Measured - % ▪ 0x05CE Low Pressure EGR Valve Position Sensor Voltage - Volts ▪ Refer to the electrical circuit diagram and check the low pressure EGR valve circuit for short circuit to ground, open circuit, high resistance

		connector pin corrosion Low pressure EGR valve failure	circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest - Check and install a new low pressure EGR valve as required - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure adaption clear' routine (0x4051_03) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0408-00	EGR Sensor "B" Circuit High - No sub type information	- EGR valve circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - EGR valve failure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x0374 EGR Valve Position Voltage - Volts 0x03FB EGR Bank 1 - Commanded - % 0xF42C Commanded EGR Valve Position - % - Refer to the electrical circuit diagram and check the EGR valve circuit for short circuit to power - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest - Check and install a new EGR valve as required - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405)

			diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure EGR valve adaption clear' routine (0x4051_03/0xF405) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0408-17	EGR Sensor "B" Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Low pressure EGR valve circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Low pressure EGR valve failure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x05CD Low Pressure EGR Valve Position - Measured - % 0x05CE Low Pressure EGR Valve Position Sensor Voltage - % - Refer to the electrical circuit diagram and check the low pressure EGR valve circuit for short circuit to power - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system, perform routine 2 Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve test the DTCs and retest - Check and install a new low pressure EGR valve as required - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure EGR valve adaption clear' routine (0x4051_03/0xF405) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

P040B-64	EGR Temperature Sensor "A" Circuit Range/Performance - Signal plausibility failure	- Exhaust gas recirculation outlet temperature sensor - High pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation outlet temperature sensor - High pressure EGR failure - Contamination - Obstruction	- Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - High pressure EGR circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. If the Exhaust gas recirculation valve solenoid is faulty, replace the solenoid. Clear the DTCs and retest - Check and install a new exhaust gas recirculation outlet temperature sensor - High pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P040B-84	EGR Temperature Sensor "A" Circuit Range/Performance - Signal below allowable range	- Exhaust gas recirculation outlet temperature sensor - High pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation outlet temperature sensor - High pressure EGR failure - Contamination - Obstruction	- Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - High pressure EGR circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. If the Exhaust gas recirculation valve solenoid is faulty, replace the solenoid. Clear the DTCs and retest - Check and install a new exhaust gas recirculation outlet temperature sensor - High pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P040B-85	EGR Temperature Sensor "A" Circuit Range/Performance	The engine control module has determined failures where some circuit quantity, reported via	Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - High

	- Signal above allowable range	serial data, is above a specified range - Exhaust gas recirculation outlet temperature sensor - High pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation outlet temperature sensor - High pressure EGR failure - Contamination - Obstruction	pressure EGR circuit for short ground, short circuit to power circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnose Exhaust gas recirculation valve the DTCs and retest - Check and install a new exhaust recirculation outlet temperature sensor - High pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P040C-16	EGR Temperature Sensor "A" Circuit Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Exhaust gas recirculation outlet temperature sensor - High pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation outlet temperature sensor - High pressure EGR failure - Contamination - Obstruction	- Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - High pressure EGR circuit for short to ground, short circuit to power circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnose Exhaust gas recirculation valve the DTCs and retest - Check and install a new exhaust recirculation outlet temperature sensor - High pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P040D-17	EGR Temperature Sensor "A" Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Exhaust gas recirculation outlet	Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - High pressure EGR circuit for short to ground, short circuit to power circuit, high resistance

		temperature sensor - High pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation outlet temperature sensor - High pressure EGR failure - Contamination - Obstruction	- Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnose Exhaust gas recirculation valve the DTCs and retest - Check and install a new exhaust recirculation outlet temperature sensor - High pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P041B-64	EGR Temperature Sensor "B" Circuit Range/Performance - Signal plausibility failure	- Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation outlet temperature sensor - Low pressure EGR failure - Contamination - Obstruction	- Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnose Exhaust gas recirculation valve the DTCs and retest - Check and install a new exhaust recirculation outlet temperature sensor - Low pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P041B-84	EGR Temperature Sensor "B" Circuit Range/Performance - Signal below allowable range	- Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation outlet	- Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion

		temperature sensor - Low pressure EGR failure - Contamination - Obstruction	- Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest - Check and install a new exhaust recirculation outlet temperature sensor - Low pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P041B-85	EGR Temperature Sensor "B" Circuit Range/Performance - Signal above allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation outlet temperature sensor - Low pressure EGR failure - Contamination - Obstruction	- Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest - Check and install a new exhaust recirculation outlet temperature sensor - Low pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P041C-16	EGR Temperature Sensor "B" Circuit Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out,	- Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic

		connector pin corrosion - Exhaust gas recirculation outlet temperature sensor - Low pressure EGR failure - Contamination - Obstruction	Exhaust gas recirculation valve the DTCs and retest - Check and install a new exhaust recirculation outlet temperature sensor - Low pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P041D-17	EGR Temperature Sensor "B" Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation outlet temperature sensor - Low pressure EGR failure - Contamination - Obstruction	- Refer to the electrical circuit diagram and check the Exhaust gas recirculation outlet temperature sensor - Low pressure EGR circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system, perform road test. If the Exhaust gas recirculation valve DTCs return, clear the DTCs and retest - Check and install a new exhaust recirculation outlet temperature sensor - Low pressure EGR as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0420-00	Catalyst System Efficiency Below Threshold Bank 1 - No Sub Type Information		- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 - Check engine control module related DTCs and refer to relevant index - Check and install a new diesel catalyst (DOC). If the DPF is below critical soot loading initiate a soot regeneration. Confirm soot level reduces correctly. If the DPF is above the critical soot loading replace and reset the soot loading valve

NOTE:

Diesel oxidation catalyst exothermic generation capability check. Diesel oxidation catalyst monitoring runs during the active regeneration of the particulate filter. The exhaust gas temperature measured at the catalyst outlet is compared against a model of the expected temperature increase during the active regeneration. A malfunction is reported when the measured temperature is significantly different to this model. The monitoring returns a judgment around 10min after the active particulate filter regeneration was started

- Other related DTCs
 - Airpath
 - Manifold
- Diesel oxidation catalyst failed, air path failures such as temperature and pressure sensors causing frequent and hot regeneration, any air path failure which will lead to increased soot loading of the DPF such as charge air leak, or blocked air filter causing frequent and hot regeneration
- Exhaust gas temperature sensor - Post catalyst circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Connector is disconnected, connector pin is backed out, connector pin corrosion
- Exhaust gas temperature sensor - Post catalyst failure

- Using the manufacturer approved diagnostic system carry out 'Oxidation catalyst' routine (0x405B_0A)
- Using the manufacturer approved diagnostic system carry out 'particulate filter dynamic regeneration' routine (0x405B_0A / DID 042C / DID F405)
- Refer to the electrical circuit diagram and check the exhaust gas temperature sensor - Post catalyst circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Inspect connectors for signs of moisture ingress, and pins for damage or corrosion
- Check and install a new exhaust gas temperature sensor - Post catalyst circuit as required
- Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P045A-13	EGR "B" Control Circuit - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Low pressure EGR valve circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Low pressure EGR valve failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals ▪ 0x05CD Low Pressure EGR Position - Measured - % ▪ 0x05CE Low Pressure EGR Position Sensor Voltage - % ▪ Refer to the electrical circuit diagram and check the low pressure EGR circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest ▪ Check and install a new low pressure EGR valve as required ▪ Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4052/0xF405) ▪ Using the manufacturer approved diagnostic system carry out 'control module low pressure adaptation clear' routine (0x4053) ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P045B-77	EGR "B" Control Circuit Range/Performance - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Low pressure EGR valve mechanical integrity ▪ Low pressure EGR valve circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals ▪ 0x05CD Low Pressure EGR Position - Measured - % ▪ 0x05CE Low Pressure EGR Position Sensor Voltage - % ▪ Check low pressure EGR valve mechanical integrity ▪ Refer to the electrical circuit diagram and check the low pressure EGR circuit for short circuit to ground

			diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure EGR valve position sensor adaption clear' routine (0x4051_03/0x4051_04/0xF405) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P045C-16	EGR "B" Control Circuit Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Low pressure EGR valve circuit short circuit to ground, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Low pressure EGR valve failure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x05CD Low Pressure EGR Valve Position - Measured - % 0x05CE Low Pressure EGR Valve Position Sensor Voltage - V - Refer to the electrical circuit diagram and check the low pressure EGR valve circuit for short circuit to ground, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Check and install a new low pressure EGR valve as required - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure EGR valve position sensor adaption clear' routine (0x4051_03/0x4051_04/0xF405) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P045D-12	EGR "B" Control Circuit High - Circuit short to battery	The engine control module has detected a vehicle power measurement for a period longer than expected or has	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x05CD Low Pressure EGR Valve Position - Measured - %

		detected a vehicle power measurement when another value was expected - Low pressure EGR valve circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Low pressure EGR valve failure	Position - Measured - % 0x05CE Low Pressure EGR Position Sensor Voltage - - Refer to the electrical circuit diagram and check the low pressure EGR circuit for short circuit to power - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest - Check and install a new low pressure EGR valve as required - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4052/0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure adaption clear' routine (0x4053) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P045D-17	EGR "B" Control Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Low pressure EGR valve circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Low pressure EGR valve failure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x05CD Low Pressure EGR Position - Measured - % 0x05CE Low Pressure EGR Position Sensor Voltage - - Refer to the electrical circuit diagram and check the low pressure EGR circuit for short circuit to power - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check and install a new low pressure EGR valve as required - Using the manufacturer approved diagnostic system carry out

			path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system carry out the control module low pressure EGR valve adaptation clear' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P045E-72	EGR "B" Control Stuck Open - Actuator stuck open	- The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to close some piece of equipment - Low pressure EGR valve mechanical integrity - Low pressure EGR valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Low pressure EGR valve failure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x05CD Low Pressure EGR Valve Position - Measured - % 0x05CE Low Pressure EGR Valve Position Sensor Voltage - V - Check low pressure EGR valve mechanical integrity - Refer to the electrical circuit diagram and check the low pressure EGR valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve test. Check the DTCs and retest - Check and install a new low pressure EGR valve as required - Using the manufacturer approved diagnostic system carry out the path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system carry out the control module low pressure EGR valve adaptation clear' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system clear all stored DTCs and retest

			using the Diagnosis Menu tab retest
P045F-73	EGR "B" Control Stuck Closed - Actuator stuck closed	▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Low pressure EGR valve mechanical integrity ▪ Low pressure EGR valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Low pressure EGR valve failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals ▪ 0x05CD Low Pressure EGR Position - Measured - % ▪ 0x05CE Low Pressure EGR Position Sensor Voltage - ▪ Refer to the electrical circuit diagram and check the low pressure EGR circuit for short circuit to ground, circuit to power, open circuit, resistance ▪ Check low pressure EGR valve mechanical integrity ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest ▪ Check and install a new low pressure EGR valve as required ▪ Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4052/0xF405) ▪ Using the manufacturer approved diagnostic system carry out 'control module low pressure adaption clear' routine (0x4053) ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab retest
P0475-13	Exhaust Pressure Control Valve "A" - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output	▪ Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion

		- Exhaust gas recirculation - Low pressure butterfly valve circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low pressure butterfly valve failure	corrosion - Check exhaust gas recirculation pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the manufacturer approved diagnostic system carry out 'Exhaust flap adaption' routine (0x0406_0D) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0476-77	Exhaust Pressure Control Valve "A" Range/Performance - Commanded position not reachable	- The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment - Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low pressure butterfly valve failure	- Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of corrosion, ingress, and pins for damage or corrosion - Check exhaust gas recirculation pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the manufacturer approved diagnostic system carry out 'Exhaust flap adaption' routine (0x0406_0D) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0477-11	Exhaust Pressure Control Valve "A" Low - Circuit short to ground	The engine control module has detected a ground measurement for a period longer than expected or has detected a	Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve for short circuit to ground, short circuit to power, open circuit, high resistance

		ground measurement when another value was expected - Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low pressure butterfly valve failure	power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check exhaust gas recirculation pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust recirculation - Low pressure butterfly valve as required - Using the manufacturer approved diagnostic system carry out 'Exhaust flap adaption' routine (0x0406_0D) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0478-12	Exhaust Pressure Control Valve "A" High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low pressure butterfly valve failure	- Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check exhaust gas recirculation pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust recirculation - Low pressure butterfly valve as required - Using the manufacturer approved diagnostic system carry out 'Exhaust flap adaption' routine (0x0406_0D) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P047F-72	Exhaust Pressure Control Valve "A" Stuck Open - Actuator stuck open	▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to close some piece of equipment ▪ Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity ▪ Exhaust gas recirculation - Low pressure butterfly valve failure	▪ Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check exhaust gas recirculation - Low pressure butterfly valve for mechanical integrity ▪ Linkage not seized ▪ Butterfly valve not stuck ▪ Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required ▪ Using the manufacturer approved diagnostic system carry out the 'Exhaust flap adaption' routine (0x0406_0D) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0480-13	Fan 1 Control Circuit - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Main electric/viscous cooling fan unable to move due to obstruction ▪ Main electric/viscous cooling fan circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Main electric/viscous cooling fan failure	NOTE: Operational requirements need to be met to allow the monitor to be fully tested. Check the vehicle at normal operating temperature until the cooling fan operates to confirm the repair ▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals ▪ 0x03B9 Engine Cooling Fan - Desired ▪ Check there is not any obstruction preventing main electric/viscous cooling fan movement ▪ Refer to the electrical circuit diagram and check the main electric/viscous cooling fan circuit for open circuit, short circuit to ground, short circuit to power, high resistance

			▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new main electric/viscous cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0480-4B	Fan 1 Control Circuit - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Main electric cooling fan unable to move due to obstruction ▪ Main electric cooling fan circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Main electric cooling fan failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signal ▪ 0x03B9 Engine Cooling Fan - Desired ▪ Check there is not any obstruction preventing main electric cooling movement ▪ Refer to the electrical circuit diagram and check the main electric cooling circuit for short circuit to power ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new main electric cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0480-71	Fan 1 Control Circuit - Actuator stuck	▪ The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay ▪ Main electric cooling fan unable to move due to obstruction ▪ Main electric cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine cooling fan stalling caused by obstruction in fan	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signal ▪ 0x03B9 Engine Cooling Fan - Desired ▪ Check there is not any obstruction preventing main electric cooling movement ▪ Refer to the electrical circuit diagram and check the main electric cooling circuit for short circuit to ground, circuit to power, open circuit, resistance ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion

		cowling ▪ Main electric cooling fan failure	corrosion ▪ Check for damage to or block fan and fouling of fan cowling as required ▪ Check and install a new main cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0480-97	Fan 1 Control Circuit - Component or system operation obstructed or blocked	▪ The engine control module has detected that the operation of a component is prevented by an obstruction ▪ Main electric cooling fan unable to move due to obstruction ▪ Main electric cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine cooling fan stalling caused by obstruction in fan cowling ▪ Main electric cooling fan failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signal ▪ 0x03B9 Engine Cooling Fan - Desired ▪ Check there is not any obstruction preventing main electric cooling fan movement ▪ Refer to the electrical circuit diagram and check the main electric cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of damage, ingress, and pins for damage and corrosion ▪ Check for damage to or block fan and fouling of fan cowling as required ▪ Check and install a new main electric cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0481-13	Fan 2 Control Circuit - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Auxiliary electric cooling fan unable to move due to obstruction	

		▪ Auxiliary electric cooling fan circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Auxiliary electric cooling fan failure	NOTE: Operational requirements need to allow the monitor to be fully tested. Check the vehicle at normal operating temperature until the cooling fan operates to confirm the repair ▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signal ▪ 0x03F9 Electric Fan PWM Commanded ▪ Check there is not any obstruction preventing auxiliary electric cooling fan movement ▪ Refer to the electrical circuit diagram and check the auxiliary electric cooling fan circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Check and install a new auxiliary electric cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0481-4B	Fan 2 Control Circuit - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Auxiliary electric cooling fan unable to move due to obstruction ▪ Auxiliary electric cooling fan circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Auxiliary electric cooling fan failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signal ▪ 0x03F9 Electric Fan PWM Commanded ▪ Check there is not any obstruction preventing auxiliary electric cooling fan movement ▪ Refer to the electrical circuit diagram and check the auxiliary electric cooling fan circuit for short circuit to power ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion

			▪ Check and install a new auxiliary cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0481-71	Fan 2 Control Circuit - Actuator stuck	▪ The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay ▪ Auxiliary electric cooling fan unable to move due to obstruction ▪ Auxiliary electric cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine cooling fan stalling caused by obstruction in fan cowling ▪ Auxiliary electric cooling fan failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signal ▪ 0x03F9 Electric Fan PWM Commanded ▪ Check there is not any obstruction preventing auxiliary electric cooling fan movement ▪ Refer to the electrical circuit diagram and check the auxiliary electric fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check for damage to or blockage of fan and fouling of fan cowling as required ▪ Check and install a new auxiliary cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0481-97	Fan 2 Control Circuit - Component or system operation obstructed or blocked	▪ The engine control module has detected that the operation of a component is prevented by an obstruction ▪ Unable to move, due to obstruction ▪ Auxiliary electric cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signal ▪ 0x03F9 Electric Fan PWM Commanded ▪ Check there is not any obstruction preventing auxiliary electric cooling fan movement ▪ Refer to the electrical circuit diagram and check the auxiliary electric fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		▪ Engine cooling fan stalling caused by obstruction in fan cowling ▪ Auxiliary electric cooling fan failure	▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check for damage to or blockage of fan and fouling of fan cowling as required ▪ Check and install a new auxiliary electric cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0489-11	EGR "A" Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ EGR valve circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ EGR valve failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals ▪ 0x0374 EGR Valve Position - Volts ▪ 0x03FB EGR Bank 1 - Commanded - % ▪ 0xF42C Commanded EGR Valve Position - % ▪ Refer to the electrical circuit diagram and check the EGR valve circuit for short circuit to ground ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest ▪ Check and install a new EGR valve as required ▪ Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4052/0xF405) ▪ Using the manufacturer approved diagnostic system carry out 'control module low pressure adaption clear' routine (0x4053) ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

P048A-73	Exhaust Pressure Control Valve "A" Stuck Closed - Actuator stuck closed	▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity ▪ Exhaust gas recirculation - Low pressure butterfly valve failure	▪ Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check exhaust gas recirculation pressure butterfly valve for mechanical integrity ▪ Linkage not seized ▪ Butterfly valve not stuck ▪ Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required ▪ Using the manufacturer approved diagnostic system carry out 'Exhaust flap adaption' routine (0x0406_0D) ▪ Using the manufacturer approved diagnostic system clear all stored faults and retest
P048D-16	Exhaust Pressure Control Valve "A" Position Sensor/Switch Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity ▪ Exhaust gas recirculation - Low pressure butterfly valve failure	▪ Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check exhaust gas recirculation pressure butterfly valve for mechanical integrity ▪ Linkage not seized ▪ Butterfly valve not stuck ▪ Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required ▪ Using the manufacturer approved diagnostic system carry out 'Exhaust flap adaption' routine (0x0406_0D)

			Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P048E-17	Exhaust Pressure Control Valve "A" Position Sensor/Switch Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Exhaust gas recirculation - Low pressure butterfly valve circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas recirculation - Low pressure butterfly valve mechanical integrity - Exhaust gas recirculation - Low pressure butterfly valve failure	- Refer to the electrical circuit diagram and check the exhaust gas recirculation - Low pressure butterfly valve for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Check exhaust gas recirculation pressure butterfly valve for mechanical integrity - Linkage not seized - Butterfly valve not stuck - Check and install a new exhaust gas recirculation - Low pressure butterfly valve as required - Using the manufacturer approved diagnostic system carry out 'Exhaust flap adaption' routine (0x0406_0D) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0490-12	EGR "A" Control Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - EGR valve circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - EGR valve failure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x0374 EGR Valve Position - Volts 0x03FB EGR Bank 1 - Commanded - % 0xF42C Commanded EGR Flow - % - Refer to the electrical circuit diagram and check the EGR valve circuit for short circuit to power - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform routine

			Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest - Check and install a new EGR valve as required - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure control module low pressure adaption clear' routine (0x4051_01/0x4051_02/0xF405) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0494-00	Fan Speed Low - No sub type information	NOTE: The sensed viscous fan speed is checked if it is within a valid minimum physical range - Main viscous cooling fan unable to move due to obstruction - Connector is disconnected, connector pin is backed out, connector pin corrosion - Main viscous cooling fan failure	- Check there is not any obstruction preventing main viscous cooling fan movement - Inspect connectors for signs of damage, ingress, and pins for damage and corrosion - Check and install a new main viscous cooling fan as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0495-00	Fan Speed High - No sub type information	NOTE: The sensed viscous fan speed is checked if it is within a valid maximum physical range - Main viscous cooling fan unable to move due to obstruction - Connector is disconnected,	- Check there is not any obstruction preventing main viscous cooling fan movement - Inspect connectors for signs of damage, ingress, and pins for damage and corrosion - Check and install a new main viscous cooling fan as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

		connector pin is backed out, connector pin corrosion ▪ Main viscous cooling fan failure	
P049B-00	EVAP "B" Flow Insufficient Detected - No sub type information	▪ Other low pressure EGR related DTCs ▪ Low pressure EGR cooler blocked / filter blocked ▪ Low pressure EGR valve mechanical integrity ▪ Low pressure EGR differential pressure sensor mechanical integrity ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check engine control module pressure EGR related DTCs and relevant DTC index ▪ Check low pressure EGR cooler blockage ▪ Check low pressure EGR valve mechanical integrity ▪ Check low pressure EGR differential pressure sensor for mechanical integrity ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P049C-00	EVAP "B" Flow Excessive Detected - No sub type information	▪ Other low pressure EGR related DTCs ▪ Low pressure EGR system leakage ▪ Low pressure EGR valve mechanical integrity ▪ Low pressure EGR differential pressure sensor mechanical integrity ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check engine control module pressure EGR related DTCs and relevant DTC index ▪ Check low pressure EGR system leakage ▪ Check low pressure EGR valve mechanical integrity ▪ Check low pressure EGR differential pressure sensor for mechanical integrity ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Using the manufacturer approved diagnostic system, perform road test. Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve the DTCs and retest ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

			using the Diagnosis Menu tab retest
P04DD-74	Cold Start EGR "A" Flow Insufficient Detected - Actuator slipping	▪ EGR valve circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas recirculation valve - High pressure ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ EGR valve failure ▪ Exhaust gas recirculation valve - High pressure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals ▪ 0x0374 EGR Valve Position Voltage - Volts ▪ 0x03FB EGR Bank 1 - Commanded - % ▪ 0xF42C Commanded EGR Valve Position - % ▪ Refer to the electrical circuit diagram and check the EGR valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas recirculation valve - High pressure ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Using the manufacturer approved diagnostic system, perform routine 'Inline diagnostic unit 2 diagnostic Exhaust gas recirculation valve position' (0x4030/0x4051_01/0x4030/0xF405) and clear the DTCs and retest ▪ Check and install a new EGR valve if required ▪ Exhaust gas recirculation valve - High pressure ▪ Using the manufacturer approved diagnostic system carry out routine 'path set-up routine' routine (0x4030/0x4051_01/0x4030/0xF405) ▪ Using the manufacturer approved diagnostic system carry out routine 'control module low pressure adaption clear' routine (0x4030/0x4051_01/0x4030/0xF405) ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P04DE-74	Cold Start EGR "A" Flow Excessive Detected - Actuator slipping	▪ EGR valve mechanical integrity ▪ EGR valve circuit short circuit to ground, short circuit to power,	▪ Check EGR valve for mechanical integrity ▪ Diagnosis of this DTC may require

		open circuit, high resistance - Exhaust gas recirculation valve - High pressure - Connector is disconnected, connector pin is backed out, connector pin corrosion - EGR valve failure - Exhaust gas recirculation valve - High pressure	the manufacturer approved diagnostic system check datalogger signals 0x0374 EGR Valve Position - Volts 0x03FB EGR Bank 1 - Commanded - % 0xF42C Commanded EGR Valve Position - % - Refer to the electrical circuit diagram and check the EGR valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Exhaust gas recirculation valve - High pressure - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system, perform road test. If the 'Inline diagnostic unit 2 diagnostic test Exhaust gas recirculation valve' routine shows the DTCs and retest - Check and install a new EGR valve if required - Exhaust gas recirculation valve - High pressure - Using the manufacturer approved diagnostic system carry out 'path set-up routine' routine (0x4030/0x4051_01/0x4052/0x4053/0xF405) - Using the manufacturer approved diagnostic system carry out 'control module low pressure adaption clear' routine (0x4054) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P04FC-64	Crankcase Ventilation System/Performance - Signal plausibility failure	- Crankcase ventilation system mechanical integrity - Disconnected crankcase vent at oil separator or air intake - Crankcase ventilation pressure sensor circuit short circuit to ground, short circuit to power,	- Check the crankcase ventilation system for mechanical integrity - Refer to the electrical circuit diagram and check the crankcase ventilation pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of

		open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Crankcase ventilation pressure sensor failure - Contamination - Obstruction - Disconnected differential pressure sensor hose - Frozen sensor	ingress, and pins for damage and corrosion - Check and install a new crankcase ventilation pressure sensor as follows: - Using the manufacturer approved diagnostic system carry out the 'control module reset crankcase ventilation pressure sensor' routine (0x0406_14) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0500-29	Vehicle Speed Sensor "A" Circuit - Signal invalid	- The value of the signal measured by the engine control module is not plausible given the operating conditions - Other vehicle speed related DTCs - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check anti-lock brake system control module for vehicle speed related DTCs and refer to relevant DTC index - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0500-84	Vehicle Speed Sensor "A" Circuit - Signal below allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range - Other vehicle speed related DTCs - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check anti-lock brake system control module for vehicle speed related DTCs and refer to relevant DTC index - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0500-85	Vehicle Speed Sensor "A" Circuit - Signal above allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Other vehicle speed related DTCs - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check anti-lock brake system control module for vehicle speed related DTCs and refer to relevant DTC index - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

P0501-29	Vehicle Speed Sensor "A" Circuit Range/Performance - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Other vehicle speed related DTCs ▪ Brake pedal switch adjustment ▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Brake pedal switch failure	▪ Check anti-lock brake system module for vehicle speed related DTCs and refer to relevant DTC index ▪ Check brake pedal switch for adjustment ▪ Refer to the electrical circuit diagram and check the brake pedal switch for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Check and install a new brake pedal switch as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0504-62	Brake Switch "A"/"B" Correlation - Signal compare failure	▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Other vehicle speed related DTCs ▪ Brake pedal switch adjustment ▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Brake pedal switch failure	▪ Check anti-lock brake system module for vehicle speed related DTCs and refer to relevant DTC index ▪ Check brake pedal switch for adjustment ▪ Refer to the electrical circuit diagram and check the brake pedal switch for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Start the engine and press the brake pedal, using maximum travel for more than 1 second, taking care not to depress the accelerator pedal at the same time ▪ Check and install a new brake pedal switch as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0506-00	Idle Control System RPM - Lower Than	▪ Other related DTCs ▪ Mechanical blockage,	▪ Check engine control module for related DTCs and refer to relevant DTC index

	Expected - No sub type information	obstruction of the FEAD ▪ Friction of crankshaft too high ▪ Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance	index ▪ Check for mechanical blockage obstruction of the FEAD ▪ Refer to the electrical circuit diagram and check connections are secure wiring integrity ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0507-00	Idle Control System RPM - Higher Than Expected - No sub type information	▪ Other related DTCs ▪ Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check engine control module related DTCs and refer to relevant index ▪ Refer to the electrical circuit diagram and check connections are secure wiring integrity ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P050E-00	Cold Start Engine Exhaust Temperature Too Low - No sub type information	NOTE: Rapid heat up system monitoring - Target temperature not met during rapid heat up. The heat generation during rapid heat up (after coldstart) is below threshold ▪ Other related DTCs ▪ Exhaust gas temperature sensor - Post catalytic converter ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post catalytic converter failure ▪ Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit,	▪ Check engine control module related DTCs and refer to relevant index ▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 ▪ 0x03E9 Exhaust Gas Temperature Bank2 Sensor 2 Voltage ▪ Refer to the electrical circuit diagram and check the exhaust gas temperature sensor - Post catalytic converter for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new exhaust temperature sensor - Post catalytic converter as required ▪ Check DPF for mechanical interference ▪ Using the manufacturer approved diagnostic system carry out

		high resistance	particulate filter dynamic regeneration' routine (0x4 DID 042C / DID F405) Refer to the electrical circuit diagram and check connections are secure and wiring integrity
P050F-00	Brake Assist Vacuum Too Low - No sub type information	- Brake vacuum switch circuit short circuit to ground, short circuit to power, open circuit, high resistance - Brake vacuum switch vacuum harness system mechanical integrity - Connector is disconnected, connector pin is backed out, connector pin corrosion - Brake vacuum switch failure - Brake vacuum switch diaphragm failure	- Refer to the electrical circuit diagram and check the brake vacuum switch circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check brake vacuum switch vacuum harness system for mechanical integrity - Inspect connectors for signs of damage, ingress, and pins for damage and corrosion - Check and install a new brake vacuum switch as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0512-12	Starter Request Circuit - Circuit short to battery	NOTE: To diagnose when the hardwired push button crank request from the central junction box is short circuit to battery - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Crank request signal from central junction box circuit to the engine control module short circuit to power - Circuit reference CRNK - Connector is disconnected,	- Refer to the electrical circuit diagram and check the crank request signal circuit from central junction box to the engine control module circuit to power - Inspect connectors for signs of damage, ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

		connector pin is backed out, connector pin corrosion	
P0512-14	Starter Request Circuit - Circuit short to ground or open	NOTE: To diagnose when the hardwired push button crank request from the central junction box is short circuit to battery - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle ground or open circuit measurement when another value was expected - Crank request signal from central junction box circuit to the engine control module short circuit to ground, open circuit, high resistance - Circuit reference CRNK - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Refer to the electrical circuit diagram and check the crank request signal circuit from central junction box to the engine control module circuit to ground, open circuit, resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0513-00	Incorrect Immobilizer Key - No sub type information	Security key invalid	- Re-configure the engine control module using the manufacturer approved diagnostic system. Re-configure instrument cluster using the manufacturer approved diagnostic system - Using the manufacturer approved diagnostic system, carry out the immobilization procedure - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P051B-64	Crankcase Pressure Sensor Circuit Range/Performance - Signal plausibility	Crankcase ventilation pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagram and check the crankcase ventilation pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance

	failure	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Crankcase ventilation pressure sensor failure ▪ Contamination ▪ Obstruction	circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new crankcase ventilation pressure sensor as follows: ▪ Using the manufacturer approved diagnostic system carry out 'control module reset crankcase ventilation pressure sensor' routine (0x0406_14) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P051B-84	Crankcase Pressure Sensor Circuit Range/Performance - Signal below allowable range	▪ Crankcase ventilation pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Crankcase ventilation pressure sensor failure ▪ Contamination ▪ Obstruction	▪ Refer to the electrical circuit diagram and check the crankcase ventilation pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new crankcase ventilation pressure sensor as follows: ▪ Using the manufacturer approved diagnostic system carry out 'control module reset crankcase ventilation pressure sensor' routine (0x0406_14) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P051C-16	Crankcase Pressure Sensor Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Crankcase ventilation pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out,	▪ Refer to the electrical circuit diagram and check the crankcase ventilation pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new crankcase ventilation pressure sensor as follows: ▪ Using the manufacturer approved diagnostic system carry out 'control module reset crankcase ventilation pressure sensor' routine (0x0406_14)

		connector pin corrosion ■ Crankcase ventilation pressure sensor failure ■ Contamination ■ Obstruction	diagnostic system carry out control module reset crankcase ventilation pressure sensor' routine (0x0406_14) ■ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P051D-17	Crankcase Pressure Sensor Circuit High - Circuit voltage above threshold	■ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ■ Crankcase ventilation pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Crankcase ventilation pressure sensor failure ■ Contamination ■ Obstruction	■ Refer to the electrical circuit diagram and check the crankcase ventilation pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ■ Check and install a new crankcase ventilation pressure sensor as required ■ Using the manufacturer approved diagnostic system carry out control module reset crankcase ventilation pressure sensor' routine (0x0406_14) ■ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0526-00	Fan Speed Sensor Circuit - No sub type information	NOTE: The signal period from viscous cooling fan feedback is checked if it is within the maximum measurement limit of the timer ■ Viscous cooling fan feedback - An input to the engine control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion	■ Refer to the electrical circuit diagram and check the viscous cooling fan feedback circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ■ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P052F-13	Glow Plug Control Module 1 System Voltage - Circuit open	NOTE: Glow plug control module T30 pin open circuit, high resistance - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - LIN_B circuit and Glow plug control module circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - LIN_B circuit or Glow plug control module failure	- Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Check and install a new glow plug control module as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P052F-16	Glow Plug Control Module 1 System Voltage - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - LIN_B circuit and Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - LIN_B circuit or Glow plug control module failure	- Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Check and install a new glow plug control module as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P052F-17	Glow Plug Control Module 1 System Voltage - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - LIN_B circuit and Glow plug control module circuit short	- Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion

		circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - LIN_B circuit or Glow plug control module failure	corrosion - Check and install a new glow control module as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0544-22	Exhaust Gas Temperature Sensor Circuit Bank 1 Sensor 1 - Signal amplitude > maximum	NOTE: To monitor for too high temperature in the turbine housing; component protection. Temperature sensor temperature above threshold (899.96degC) - The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high - Other related DTCs - Airpath - Turbocharger - EGR - Fuel system - Turbocharger housing mechanical integrity - Fuel injector (s) mechanical integrity - Cylinders 1, 2 and 3 - Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Turbine temperature sensor failure	- Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signals 0x05CA Exhaust Turbine A Temperature - Check temperature sensor plausibility with engine off 0x05C9 Exhaust Turbine A Temperature Sensor Voltage - Check sensor for voltage - Check engine control module related DTCs and refer to relevant index - Check turbocharger housing for mechanical integrity - Check fuel injector (s) for mechanical integrity - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check and install a new exhaust temperature sensor - Turbine temperature sensor as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

P0545-16	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Turbine temperature sensor failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signal ▪ 0x05C9 Exhaust Turbine A Temperature Sensor Voltage ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new exhaust temperature sensor - Turbine temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0546-17	Exhaust Gas Temperature Sensor Circuit High Bank 1 Sensor 1 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Turbine temperature sensor failure	▪ Diagnosis of this DTC may require the manufacturer approved diagnostic system check datalogger signal ▪ 0x05C9 Exhaust Turbine A Temperature Sensor Voltage ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new exhaust temperature sensor - Turbine temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P054E-00	Idle Control System - Fuel Quantity Lower Than Expected - No sub type information	▪ Fuel injector (s) mechanical integrity ▪ Crankshaft friction excessive ▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Engine control module mechanical integrity	▪ Check fuel injector (s) for mechanical integrity ▪ Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Check engine control module

			mechanical integrity - Check and install a new engine module as required - Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'fuel pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P054F-00	Idle Control System - Fuel Quantity Higher Than Expected - No sub type information	- Fuel injector (s) mechanical integrity - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Check fuel injector (s) for mechanical integrity - Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine module as required - Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'fuel pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0555-29	Brake Booster Pressure Sensor Circuit - Signal invalid	- The value of the signal measured by the engine control module is not plausible given the operating conditions - CAN signal error for brake	- Check anti-lock brake system module for related DTCs and relevant DTC index - Using the manufacturer approved diagnostic system, perform a

		pressure over CAN bus	network integrity test - Refer to the electrical circuit diagram and check anti-lock brake system control module power and ground connections for open circuit - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0562-00	System Voltage Low - No sub type information	- Vehicle battery under charged - Charging system fault - under charging - Generator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Excessive quiescent current	- Refer to the workshop manual battery care manual, inspect the battery and ensure it is fully charged and serviceable before performing further tests - Refer to the electrical circuit diagram and check the power and ground connections to the module - Check the vehicle charging system performance to ensure the voltage regulation is correct - Refer to the electrical circuit diagram and check the generator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0562-16	System Voltage Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Other related DTCs - Vehicle battery under charged - Charging system fault - under charging - Power and ground connections to the engine control module, open circuit, high resistance	- Check engine control module related DTCs and refer to relevant index - Refer to the workshop manual battery care manual, inspect the battery and ensure it is fully charged and serviceable before performing further tests - Refer to the electrical circuit diagram and check the power and ground connections to the module - Refer to the electrical circuit diagram and check the generator circuit

		- Generator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check the vehicle charging system performance to ensure the voltage regulation is correct - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0563-00	System Voltage High - No sub type information	- Other related DTCs - Vehicle battery over charged - Charging system fault - over charging - Generator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module main fuse mechanical integrity - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module related DTCs and refer to relevant index - Check the vehicle charging system performance to ensure the voltage regulation is correct - Refer to the electrical circuit diagram and check the generator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Check engine control module for mechanical integrity - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0563-17	System Voltage High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Other related DTCs - Vehicle battery over charged - Charging system fault - over charging - Generator circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Check engine control module related DTCs and refer to relevant index - Check the vehicle charging system performance to ensure the voltage regulation is correct - Refer to the electrical circuit diagram and check the generator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion

		Connector is disconnected, connector pin is backed out, connector pin corrosion	Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0571-62	Brake Switch "A" Circuit - Signal compare failure	- The engine control module detected failure when comparing two or more input parameters for plausibility - Other vehicle speed related DTCs - Brake pedal switch adjustment - Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Brake pedal switch failure	- Check anti-lock brake system module for vehicle speed related DTCs and refer to relevant DTC index - Check brake pedal switch for adjustment - Refer to the electrical circuit diagram and check the brake pedal switch for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Check and install a new brake pedal switch as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0575-29	Cruise Control Input Circuit - Signal invalid	- The value of the signal measured by the engine control module is not plausible given the operating conditions - Other speed control system failure related DTCs - Other communications bus network failure related DTCs - Speed control buttons jammed/contaminated/damaged - Clockspring mechanical integrity - Speed control system failure connector is disconnected, connector pin is backed out, connector pin corrosion - Speed control module power or ground circuit short circuit to ground, circuit short circuit to power	- Check speed control module for DTCs and refer to relevant DTC index - Check communications bus network related DTCs and refer to relevant DTC index - Check speed control buttons for jammed/contaminated/damaged. Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Check clockspring for mechanical integrity - Refer to the electrical circuit diagram and check speed control module and ground circuits for short circuit to ground, circuit short circuit to power - Using the manufacturer approved diagnostic system, perform a network integrity test - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

			retest
P0575-81	Cruise Control Input Circuit - Invalid serial data received	▪ The engine control module has indicated a signal was received with the corresponding validity bit equal to "invalid" or post processing of the signal determines it is invalid ▪ Other speed control system failure related DTCs ▪ Other communications bus network failure related DTCs ▪ Speed control buttons jammed/contaminated/damaged ▪ Clockspring mechanical integrity ▪ Speed control system failure connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Speed control module power or ground circuit short circuit to ground, circuit short circuit to power	▪ Check speed control module for DTCs and refer to relevant DTC index ▪ Check communications bus network related DTCs and refer to relevant index ▪ Check speed control buttons for jammed/contaminated/damaged. Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check clockspring for mechanical integrity ▪ Refer to the electrical circuit diagram and check speed control module and ground circuits for short circuit to ground, circuit short circuit to power ▪ Using the manufacturer approved diagnostic system, perform a network integrity test ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P059F-07	Active Grille Air Shutter "A" Performance/Stuck Off - Mechanical failures	▪ Debris preventing operation of active grille air shutter and active grille air shutter motor ▪ Active grille air shutter has become disconnected from the active grille air shutter motor ▪ Linkage between the active grille air shutter and active grille air shutter motor has failed	▪ Check the active grille air shutter and active grille air shutter motor for mechanical failure ▪ Check and install a new active grille air shutter and active grille air shutter motor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P059F-13	Active Grille Air Shutter "A" Performance/Stuck Off - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ The engine control module has not received the expected signal from the active grille air shutter	▪ Check Active grille air shutter for mechanical integrity ▪ Refer to the electrical circuit diagram and check the active grille air shutter circuit for open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion

		actuator - Active grille air shutter mechanical integrity - Active grille air shutter circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Active grille air shutter failure	- Check and install a new active shutter as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P05A2-01	Active Grille Air Shutter "A" Control Circuit/Open - General electrical failure	- Active grille air shutter LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Active grille air shutter failure	- Refer to the electrical circuit diagram and check the active grille air shutter LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Check and install a new active shutter as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P05A2-97	Active Grille Air Shutter "A" Control Circuit/Open - Component or system operation obstructed or blocked	- The engine control module has detected that the operation of a component is prevented by an obstruction - Other active grille air shutter and active grille air shutter motor related DTCs - Debris preventing operation of active grille air shutter and active grille air shutter motor - Active grille air shutter has become disconnected from the active grille air shutter motor - Linkage between the active grille air shutter and active grille air shutter motor has failed	- Check engine control module for active grille air shutter and active grille air shutter motor related DTCs and refer to relevant DTC index - Check the active grille air shutter and active grille air shutter motor for mechanical failure - Check and install a new active grille air shutter and active grille air shutter motor as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P05A6-01	Active Grille Air Shutter "A" Supply	Active grille air shutter and active grille air shutter motor	Refer to the electrical circuit diagram and check the active grille air

	Voltage Circuit/Open - General electrical failure	circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Active grille air shutter and active grille air shutter motor failure	and active grille air shutter motor for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check and install a new active grille air shutter and active grille air shutter motor as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P05C0-4B	Active Grille Air Shutter Module "A" Over Temperature - Over temperature	- The engine control module detected an internal temperature above the expected range - Active grille air shutter and active grille air shutter motor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Active grille air shutter and active grille air shutter motor mechanical integrity - Active grille air shutter and active grille air shutter motor failure	- Refer to the electrical circuit diagram and check the active grille air shutter and active grille air shutter motor for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check active grille air shutter and active grille air shutter motor for mechanical integrity - Check and install a new active grille air shutter and active grille air shutter motor as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P05ED-16	Reductant Heater Control Module Supply Voltage - Circuit voltage below threshold	NOTE: To monitor diesel exhaust fluid heater control unit. Diesel exhaust fluid heater control unit under voltage - Purpose of the DTC - To monitor diesel exhaust fluid heater control unit - Electrical Cause - Yes	NOTE: This DTC when logged with intermittent status on its own is advisory only and no further action should be taken - Using the manufacturer approved diagnostic system, check for related DTCs - If this DTC exists with any other exhaust fluid DTCs, follow the

		■ Mechanical Cause ■ No ■ Control Module Cavity ■ Circuit reference B_D_CANH3 ■ Circuit reference B_D_CANL3 ■ Monitor Description ■ Diesel exhaust fluid heater control unit under voltage ■ Prioritised List of Possible Causes ■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ Diesel exhaust fluid heater control unit power and ground circuits open circuit ■ Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Diesel exhaust fluid heater control unit failure	given for that DTC ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest ■ Vehicle Conditions to enable DTC Logging strategy ■ Engine running ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check diesel exhaust fluid heater control unit power and ground circuits for open circuit ■ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for bent and/or corrosion ■ Check and install a new diesel exhaust fluid heater control unit if required ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu and retest
P05ED-17	Reductant Heater Control Module Supply Voltage - Circuit voltage above threshold	■ Purpose of the DTC ■ To monitor diesel exhaust fluid heater control unit ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ No ■ Control Module Cavity ■ Circuit reference B_D_CANH3 ■ Circuit reference B_D_CANL3 ■ Monitor Description ■ Diesel exhaust fluid heater control unit over voltage	■ Vehicle Conditions to enable DTC Logging strategy ■ Engine running ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check diesel exhaust fluid heater control unit power and ground circuits for short circuit to ground, circuit short circuit to power ■ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to ground, short circuit to power ■ inspect connectors for signs of water ingress, and pins for bent and/or corrosion

		▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Diesel exhaust fluid heater control unit power or ground circuit short circuit to ground, circuit short circuit to power ▪ Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	water ingress, and pins for and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit if required ▪ Using the manufacturer approved diagnostic system clear all DTCs using the Diagnosis and retest
P05F1-73	Reductant Pump Supply Voltage Circuit Stuck On - Actuator stuck closed	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid relay ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid relay stuck closed ▪ Prioritised List of Possible Causes ▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for and/or corrosion ▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all DTCs using the 'Diagnosis

		resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid relay failure	tab and retest
P05F2-72	Reductant Pump Supply Voltage Circuit Stuck Off - Actuator stuck open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid relay ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid relay stuck open ▪ Prioritised List of Possible Causes ▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to close some piece of equipment ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid relay	▪ Vehicle Conditions to enable Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for bent and/or corrosion ▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all DTCs using the 'Diagnosis' tab and retest

		failure	
P05F8-13	Reductant Heater Control Module Performance - Circuit open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference B_D_CANH3 ▪ Circuit reference B_D_CANL3 ▪ Monitor Description ▪ Diesel exhaust fluid heater control unit missing voltage ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid heater control unit power or ground circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	▪ Vehicle Conditions to enable Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for bent and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit if required ▪ Using the manufacturer approved diagnostic system clear all DTCs using the Diagnosis and retest
P05F8-4B	Reductant Heater Control Module Performance - Over temperature	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference	▪ Vehicle Conditions to enable Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control circuit for short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for bent and/or corrosion

		B_D_CANH3 ▪ Circuit reference B_D_CANL3 ▪ Monitor Description ▪ Diesel exhaust fluid heater control unit over temperature ▪ Prioritised List of Possible Causes ▪ The engine control module detected an internal temperature above the expected range ▪ Diesel exhaust fluid heater control unit circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control module as required ▪ Using the manufacturer approved diagnostic system clear all DTCs using the Diagnosis and retest
P05F8-66	Reductant Heater Control Module Performance - Signal has too many transitions / events	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control module cavity ▪ Circuit reference B_D_CANH3 ▪ Circuit reference B_D_CANL3 ▪ Monitor Description ▪ Diesel exhaust fluid heater control unit non volatile memory error ▪ Prioritised List of Possible Causes ▪ The engine control module monitored a parameter over time within specified limits and detected more than the expected number of transitions	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control circuit for short circuit to ground short circuit to power, open circuit high resistance ▪ Inspect connectors for signs of water ingress, and pins for bent and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control module as required ▪ Using the manufacturer approved diagnostic system clear all DTCs using the Diagnosis and retest

		▪ Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	
P05FF-00	Brake Pressure Sensor / Brake Pedal Position Sensor Correlation - No sub type information	▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Brake pedal switch failure	▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new brake switch as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P05FF-84	Brake Pressure Sensor / Brake Pedal Position Sensor Correlation - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Brake pedal switch failure	▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new brake switch as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P05FF-85	Brake Pressure Sensor / Brake Pedal Position Sensor Correlation - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new brake switch as required

		power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Brake pedal switch failure	switch as required Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0602-00	Control Module Programming Error - No sub type information	- Unapproved aftermarket modification - Mismatch between car configuration file and engine control module calibration for expected engine power output - Incorrect engine control module calibration for vehicle specification - Corrupt engine control module software download	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0605-47	Internal Control Module Read Only Memory (ROM) Error - Watchdog / safety microController failure	- Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060A-00	Internal Control Module Monitoring	Engine control module software calibration error	Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module

	Processor Performance - No sub type information	- Engine control module connector is disconnected, connector pin is backed out, connector pin corrosion - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	latest relevant level of software engine control module - Inspect connectors for signs of ingress, and pins for damage or corrosion - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'fuel pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P060A-02	Internal Control Module Monitoring Processor Performance - General signal failure	- Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance	- Using the manufacturer approved diagnostic system check and install latest relevant level of software engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P060A-16	Internal Control Module Monitoring Processor Performance - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance	- Using the manufacturer approved diagnostic system check and install latest relevant level of software engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

			retest
P060A-17	Internal Control Module Monitoring Processor Performance - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060A-29	Internal Control Module Monitoring Processor Performance - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060A-46	Internal Control Module Monitoring Processor Performance - Calibration / parameter memory failure	▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060A-49	Internal Control Module Monitoring Processor Performance - Internal electronic	▪ The engine control module detected an internal circuit failure ▪ Engine control module software calibration error	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram

	failure	Engine control module power and ground circuits open circuit, high resistance	and check engine control module power and ground circuits for circuit, high resistance Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060A-63	Internal Control Module Monitoring Processor Performance - Circuit / component protection time-out	- The engine control module detected a function is active for greater than a specified time period - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance	- Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for circuit, high resistance - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060A-83	Internal Control Module Monitoring Processor Performance - Value of signal protection calculation incorrect	- The engine control module has indicated that a message was processed with an incorrect protection (checksum) calculation - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance	- Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for circuit, high resistance - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060B-00	Internal Control Module A/D Processing Performance - No sub type information	- Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance	- Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for circuit, high resistance - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

P060B-16	Internal Control Module A/D Processing Performance - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060B-17	Internal Control Module A/D Processing Performance - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060B-1C	Internal Control Module A/D Processing Performance - Circuit voltage out of range	▪ The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low ▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P060B-29	Internal Control Module A/D Processing Performance - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Engine control module software calibration error	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance

		Engine control module power and ground circuits open circuit, high resistance	power and ground circuits for circuit, high resistance Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P060B-63	Internal Control Module A/D Processing Performance - Circuit / component protection time-out	- The engine control module detected a function is active for greater than a specified time period - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software for the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P060C-00	Internal Control Module Main Processor Performance - No sub type information	- Serial Peripheral Interface (SPI) communication bus disrupted, SPI bus is overloaded by short circuit - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software for the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P060D-29	Internal Control Module Accelerator	The value of the signal measured by the engine control module is	Using the manufacturer approved diagnostic system check and install the latest relevant level of software for the engine control module

	Pedal Position Performance - Signal invalid	not plausible given the operating conditions - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	latest relevant level of software engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out 'Engine Control Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'Fuel pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0611-16	Fuel Injector Control Module Performance - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Other related DTCs - Vehicle battery under charged - Charging system fault - under charging - Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module related DTCs and refer to relevant index - Refer to the workshop manual battery care manual, inspect the battery and ensure it is fully charged and serviceable before performing further tests - Refer to the electrical circuit diagram and check the power and ground connections to the module - Check the vehicle charging system performance to ensure the voltage regulation is correct - Refer to the electrical circuit diagram and check the fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P0611-17	Fuel Injector Control Module Performance - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Other related DTCs ▪ Vehicle battery over charged ▪ Charging system fault - over charging ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check engine control module related DTCs and refer to rele index ▪ Check the vehicle charging sy performance to ensure the vo regulation is correct ▪ Refer to the electrical circuit d and check the fuel injector cir short circuit to ground, short c power, open circuit, high resis ▪ Inspect connectors for signs o ingress, and pins for damage i corrosion ▪ Using the manufacturer appro diagnostic system clear all sto using the Diagnosis Menu tab retest
P0615-04	Starter Relay "A" Circuit - System internal failures	NOTE: To diagnose internal engine control module failure of the starter control powerstage which may inhibit starting ▪ Other related DTCs ▪ Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Starter motor relay failure ▪ Engine control module failure	▪ Check engine control module related DTCs and refer to rele index ▪ Refer to the electrical circuit d and check the starter motor re for short circuit to ground, sho to power, open circuit, high re ▪ Inspect connectors for signs o ingress, and pins for damage i corrosion ▪ Check and install a new starte relay as required ▪ Check and install a new engin module as required ▪ Using the manufacturer ap diagnostic system carry o Module - Powertrain cont module - Diesel' routine ▪ Using the manufacturer ap diagnostic system carry o pump' routine (0x4027_00 ▪ Using the manufacturer appro diagnostic system clear all sto using the Diagnosis Menu tab retest

P0615-13	Starter Relay "A" Circuit - Circuit open	NOTE: To diagnose an open load condition in the starter motor relay circuits - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Starter motor relay circuit short circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Starter motor relay failure	- Refer to the electrical circuit diagram and check the starter motor relay for open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Check and install a new starter relay as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0615-4B	Starter Relay "A" Circuit - Over temperature	NOTE: To diagnose an over temperature condition in the engine control module starter relays powerstage - The engine control module detected an internal temperature above the expected range - Other related DTCs - The engine control module detected an internal temperature above the expected range - Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check the starter motor relay for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Check and install a new starter relay as required - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'oil pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

		▪ Starter motor relay failure ▪ Engine control module failure	
P0616-00	Starter Relay "A" Circuit Low - No sub type information	▪ Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Starter motor relay failure	▪ Refer to the electrical circuit diagram and check the starter motor relay for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of damage, ingress, and pins for damage and corrosion ▪ Check and install a new starter relay as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0616-11	Starter Relay "A" Circuit Low - Circuit short to ground	NOTE: To diagnose a short circuit to ground on the high side of the starter motor control relay. To diagnose a short circuit to ground on the low side of the starter motor control relay ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Starter motor relay circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Starter motor relay failure	▪ Refer to the electrical circuit diagram and check the starter motor relay for short circuit to ground ▪ Inspect connectors for signs of damage, ingress, and pins for damage and corrosion ▪ Check and install a new starter relay as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0616-16	Starter Relay "A" Circuit Low - Circuit voltage below threshold		▪ Check engine control module related DTCs and refer to relevant index ▪ Refer to the electrical circuit diagram

		NOTE: To diagnose an internal failure with the high side ADC (analog-to-digital converter) return signal ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Other related DTCs ▪ Starter motor relay circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Starter motor relay failure ▪ Engine control module failure	and check the starter motor relay for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new starter relay as required ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0617-00	Starter Relay "A" Circuit High - No sub type information	▪ Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Starter motor relay failure	▪ Refer to the electrical circuit diagram and check the starter motor relay for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new starter relay as required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0617-12	Starter Relay "A" Circuit High - Circuit short to battery		▪ Refer to the electrical circuit diagram and check the starter motor relay for short circuit to power ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new starter relay as required

		NOTE: To diagnose a short circuit to power on the high side of the starter motor control relay. To diagnose a short circuit to power on the low side of the starter motor control relay ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Starter motor relay circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Starter motor relay failure	▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0617-17	Starter Relay "A" Circuit High - Circuit voltage above threshold	NOTE: To diagnose an internal failure with the high side ADC (analog-to-digital converter) return signal ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Other related DTCs ▪ Starter motor relay circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Starter motor relay failure	▪ Check engine control module related DTCs and refer to relevant index ▪ Refer to the electrical circuit diagram and check the starter motor relay for short circuit to power ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new starter relay as required ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Engine Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'Engine pump' routine (0x4027_00) ▪ Using the manufacturer approved

		Engine control module failure	diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061A-49	Internal Control Module Torque Performance - Internal electronic failure	- The engine control module detected an internal circuit failure - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061A-61	Internal Control Module Torque Performance - Signal calculation failure	- Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'pump' routine (0x4027_00)

			Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P061B-21	Internal Control Module Torque Calculation Performance - Signal amplitude < minimum	NOTE: The desired torque request from the vehicle supervisory controller exceeds the minimum permitted torque of monitoring functions torque distribution - The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low - Vehicle supervisory controller torque failure (software contained within the engine control module for control of the hybrid electric vehicle system) - Other HEV system failures - Corrupt engine control module software - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check engine control module for additional DTCs and refer to the DTC index. Rectify these first - Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for correct circuit, high resistance - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'fuel pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P061B-22	Internal Control Module Torque Calculation Performance - Signal amplitude > maximum	NOTE: The desired torque request from the vehicle supervisory controller exceeds the maximum permitted torque of monitoring functions torque distribution	- Check engine control module for additional DTCs and refer to the DTC index. Rectify these first - Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for correct circuit, high resistance

		▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Vehicle supervisory controller torque failure (software contained within the engine control module for control of the HEV system) ▪ Other HEV system failures ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out the 'Engine Control Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out the 'Engine control pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061B-2B	Internal Control Module Torque Calculation Performance - Signal cross coupled	NOTE: The vehicle supervisory controller electric machine torque request exceeds the electric machine generating torque capability at low shaft rotation speed ▪ Other HEV system failures ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check engine control module for additional DTCs and refer to the DTC index. Rectify these first ▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for correct wiring, correct circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out the 'Engine Control Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out the 'Engine control pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061B-35	Internal Control Module Torque		▪ Check transmission control module for additional DTCs and refer to the DTC index

	Calculation Performance - Signal high time > maximum	NOTE: The K0 clutch status does not reflect the electric machine's to CM speed ratio ▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Other transmission system failures ▪ Corrupt transmission control module software ▪ Transmission control module power supply failure ▪ Transmission control module ground supply failure ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	DTC index. Rectify these first ▪ Using the manufacturer approved diagnostic system, re-configure transmission control module with latest level software ▪ Refer to the electrical circuit diagram and check the transmission control module power and ground circuit for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, re-configure engine control module with the latest level software ▪ Refer to the electrical circuit diagram and check the engine control module power and ground circuits for open circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061B-62	Internal Control Module Torque Calculation Performance - Signal compare failure	NOTE: The vehicle supervisory controller torque request exceeds the permitted torque during K0 engine speed synchronisation ▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Speed synchronisation failure, set by engine control module torque calculation error	▪ Check engine control module for additional DTCs and refer to the DTC index. Rectify these first ▪ Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine

		▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061B-64	Internal Control Module Torque Calculation Performance - Signal plausibility failure	NOTE: The vehicle supervisory controller torque request exceeds the permitted torque during K0 engine speed synchronisation ▪ The engine control module detected plausibility failures ▪ Speed synchronisation failure, set by engine control module torque calculation error ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check engine control module for additional DTCs and refer to the DTC index. Rectify these first ▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for correct circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061B-74	Internal Control Module Torque Calculation Performance - Actuator slipping	NOTE: Diagnostic fault check to report the slip start error ▪ Other HEV system failures ▪ Corrupt engine control module software ▪ Engine control module power supply failure	▪ Check engine control module for additional DTCs and refer to the DTC index. Rectify these first ▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for correct circuit, high resistance ▪ Check and install a new engine control module as required

		▪ Engine control module ground supply failure ▪ Engine control module failure	module as required ▪ Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'oil pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061B-78	Internal Control Module Torque Calculation Performance - Alignment or adjustment incorrect	NOTE: The vehicle supervisory controller torque request exceeds the permitted torque during K0 engine speed synchronisation ▪ The engine control module has detected incorrectly adjusted or aligned components ▪ Speed synchronisation failure, set by engine control module torque calculation error ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for correct circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'oil pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061B-84	Internal Control Module Torque Calculation Performance - Signal below allowable range		▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for correct circuit, high resistance

		NOTE: The vehicle supervisory controller electric machine's torque request exceeds the electric machine's generating torque capability ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Other HEV system failures ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	power and ground circuits for circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out the 'engine pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061B-85	Internal Control Module Torque Calculation Performance - Signal above allowable range	NOTE: The vehicle supervisory controller electric machine's torque request exceeds the electric machine's motoring torque capability ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Other HEV system failures ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground	▪ Check engine control module for additional DTCs and refer to the DTC index. Rectify these first ▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out the 'engine pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults

		supply failure Engine control module failure	using the Diagnosis Menu tab retest
P061B-92	Internal Control Module Torque Calculation Performance - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab retest
P061B-94	Internal Control Module Torque Calculation Performance - Unexpected operation	- The engine control module has detected that the component is operating in a way or at a time that it has not been commanded to operate - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'pump' routine (0x4027_00)

			Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061B-9A	Internal Control Module Torque Calculation Performance - Component or system operating conditions	- Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P061C-29	Internal Control Module Engine RPM Signal invalid	- The value of the signal measured by the engine control module is not plausible given the operating conditions - Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

			pump' routine (0x4027_00) Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0627-13	Fuel Pump "A" Control Circuit / Open - Circuit open	- The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Low fuel level condition - Fuel pump - Low pressure output command from engine control module to central junction box circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel pump - Low pressure output command from engine control module to central junction box circuit failure	- Check the fuel level, add fuel - Refer to the electrical circuit diagram and check the fuel pump - Low pressure output command from engine control module to central junction box circuit for open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0628-11	Fuel Pump "A" Control Circuit Low - Circuit short to ground	- The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Low fuel level condition - Fuel pump - Low pressure output command from engine control module to central junction box circuit to ground - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel pump - Low pressure output command from engine control module to central junction box circuit failure	- Check the fuel level, add fuel - Refer to the electrical circuit diagram and check the fuel pump - Low pressure output command from engine control module to central junction box circuit for short circuit to ground - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0629-	Fuel Pump "A"	The engine control module has	Check the fuel level, add fuel

12	Control Circuit High - Circuit short to battery	detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Low fuel level condition - Fuel pump - Low pressure output command from engine control module to central junction box circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel pump - Low pressure output command from engine control module to central junction box circuit failure	- Refer to the electrical circuit diagram and check the fuel pump - Low pressure output command from engine control module to central junction box circuit for short circuit to power - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P062A-92	Fuel Pump "A" Control Circuit Range/Performance - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - Low fuel level condition - Fuel pump - Low pressure feedback circuit to engine control module circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel pump - Low pressure feedback circuit to engine control module circuit failure	- Check the fuel level, add fuel - Refer to the electrical circuit diagram and check the fuel pump - Low pressure feedback circuit to engine control module circuit for open circuit or high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P062A-98	Fuel Pump "A" Control Circuit Range/Performance - Component or system over temperature	- The engine control module has detected that the temperature is too high for the correct operation of the component or system - Low fuel level condition - Fuel pump - Low pressure output command from engine control module to central junction box circuit to power	- Check the fuel level, add fuel - Refer to the electrical circuit diagram and check the fuel pump - Low pressure output command from engine control module to central junction box circuit for short circuit to power - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel pump - Low pressure output command from engine control module to central junction box circuit failure	diagnostic system clear all sto using the Diagnosis Menu tab retest
P062B-00	Internal Control Module Fuel Injector Control Performance - No sub type information	▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Engine control module mechanical integrity	▪ Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Check engine control module mechanical integrity ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Engine Control Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'Fuel pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab retest
P062B-41	Internal Control Module Fuel Injector Control Performance - General checksum failure	▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Engine control module mechanical integrity	▪ Using the manufacturer approved diagnostic system check and install latest relevant level of software on engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Check engine control module mechanical integrity ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Engine Control Module - Powertrain control module - Diesel' routine

			module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'Fuel pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P062B-48	Internal Control Module Fuel Injector Control Performance - Supervision software failure	- Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'Fuel pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P062B-62	Internal Control Module Fuel Injector Control Performance - Signal compare failure	- Engine control module software calibration error - Engine control module power and ground circuits open circuit, high resistance - Engine control module mechanical integrity	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module - Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance - Check engine control module mechanical integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'Fuel pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

			diagnostic system carry out Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P062B-64	Internal Control Module Fuel Injector Control Performance - Signal plausibility failure	▪ The engine control module detected plausibility failures ▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Engine control module mechanical integrity	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Check engine control module mechanical integrity ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P062B-91	Internal Control Module Fuel Injector Control Performance - Parametric	▪ The engine control module has detected that the component parameter e.g. capacitance or inductance is outside its expected range ▪ Engine control module software calibration error ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Engine control module	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for open circuit, high resistance ▪ Check engine control module mechanical integrity ▪ Check and install a new engine control module as required

		mechanical integrity	module as required - Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P062D-11	Fuel Injector Driver Circuit Performance Bank 1 - Circuit short to ground	- The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - All fuel injector(s) circuit short circuit to ground, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Harness failure - Wiring integrity all fuel injector(s) circuit short circuit to power, open circuit, high resistance - Other electronic engine control related DTCs - Engine control module power and ground circuits open circuit, high resistance - Corrupt engine control module software - Engine control module failure	- Refer to the electrical circuit diagram and check all fuel injector(s) circuit for short circuit to ground, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Check and install a new harness if required - Check engine control module electronic engine control related DTCs and refer to relevant DTC index - Refer to the electrical circuit diagram and check engine control module power and ground circuits for short circuit, high resistance - Using the manufacturer approved diagnostic system, re-configure engine control module with the latest level software - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P062D-17	Fuel Injector Driver Circuit Performance Bank 1 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ All fuel injector(s) circuit short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity all fuel injector(s) circuit short circuit to power, open circuit, high resistance ▪ Other electronic engine control related DTCs ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Corrupt engine control module software ▪ Engine control module failure	▪ Refer to the electrical circuit diagram and check all fuel injector(s) circuit for short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new harness if required ▪ Check engine control module electronic engine control related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for short circuit, high resistance ▪ Using the manufacturer approved diagnostic system, re-configure engine control module with the latest level software ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'Injection pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P062E-11	Fuel Injector Driver Circuit Performance Bank 2 - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ All fuel injector(s) circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to the electrical circuit diagram and check all fuel injector(s) circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new harness if required ▪ Check engine control module electronic engine control related DTCs and refer to relevant DTC index

		▪ Harness failure - Wiring integrity all fuel injector(s) circuit short circuit to power, open circuit, high resistance ▪ Other electronic engine control related DTCs ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Corrupt engine control module software ▪ Engine control module failure	▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for short circuit, high resistance ▪ Using the manufacturer approved diagnostic system, re-configure engine control module with the latest level software ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Engine Control Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'Fuel pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P062E-17	Fuel Injector Driver Circuit Performance Bank 2 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ All fuel injector(s) circuit short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity all fuel injector(s) circuit short circuit to power, open circuit, high resistance ▪ Other electronic engine control related DTCs ▪ Engine control module power and ground circuits open circuit, high resistance ▪ Corrupt engine control module software ▪ Engine control module failure	▪ Refer to the electrical circuit diagram and check all fuel injector(s) circuit for short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new harness if required ▪ Check engine control module for electronic engine control related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagram and check engine control module power and ground circuits for short circuit, high resistance ▪ Using the manufacturer approved diagnostic system, re-configure engine control module with the latest level software ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out

			Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out the 'pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P062F-44	Internal Control Module EEPROM Error - Data Memory Failure	▪ Engine control module software error ▪ Charging system fault - over charging ▪ Engine control module to harness connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ Check the vehicle charging system performance to ensure the voltage regulation is correct ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0633-00	Immobilizer Key Not Programmed - ECM/PCM - No sub type information	▪ Power is lost from the engine control module or the central junction box during the immobilizer learn routine ▪ Security key invalid	▪ Refer to the electrical circuit diagram and check the power and ground connections to the engine control module and central junction box ▪ Re-configure the engine control module using the manufacturer approved diagnostic system. Re-configure the instrument cluster using the manufacturer approved diagnostic system ▪ Using the manufacturer approved diagnostic system, carry out the immobilization procedure ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0634-85	Control Module Internal Temperature "A"	▪ The engine control module has determined failures where some circuit quantity, reported via	▪ Check engine control module related DTCs and refer to relevant index

	Too High - Signal above allowable range	serial data, is above a specified range - Other related DTCs - Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module internal temperature sensor failure	- Refer to the electrical circuit diagram and check connections are secure and wiring integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the 'Engine Control Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the 'Engine control pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0642-00	Sensor Reference Voltage "A" Circuit Low - No sub type information	- Other sensor related DTCs 5 volt sensor circuit short circuit to ground, open circuit, high resistance - EGR valve - High pressure - EGR valve - Low pressure - Manifold absolute pressure sensor and temperature sensor - Electric throttle - Charge air pressure sensor - Variable geometry turbocharger - Primary - Accelerator pedal position sensor - Crankshaft position sensor - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check the 5 volt sensor circuit short circuit to ground, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0643-00	Sensor Reference Voltage "A" Circuit High - No sub type information	- Other sensor related DTCs 5 volt sensor circuit short circuit to ground, open circuit, high resistance - EGR valve - High pressure - EGR valve - Low pressure - Manifold absolute pressure	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check the 5 volt sensor circuit short circuit to ground, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion

		- and temperature sensor ▪ Electric throttle ▪ Charge air pressure sensor ▪ Variable geometry turbocharger - Primary ▪ Accelerator pedal position sensor ▪ Crankshaft position sensor ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	ingress, and pins for damage corrosion ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P064C-16	Glow Plug Control Module 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ LIN_B circuit and Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ LIN_B circuit or Glow plug control module failure	▪ Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of damage, ingress, and pins for damage corrosion ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P064C-17	Glow Plug Control Module 1 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ LIN_B circuit and Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ LIN_B circuit or Glow plug control module failure	▪ Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of damage, ingress, and pins for damage corrosion ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest

P064C-96	Glow Plug Control Module 1 - Component internal failure	▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ▪ LIN_B circuit or Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ LIN_B circuit or Glow plug control module failure	▪ Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for circuit to ground, short circuit, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P064D-00	Internal Control Module O2 Sensor Processor Performance Bank 1 - No sub type information	NOTE: SPI communication within heated oxygen sensor internal chip has been disrupted ▪ Other charged air system related DTCs ▪ Other fuel system related DTCs ▪ Engine control module failure	▪ Check engine control module charged air system related DTCs refer to relevant DTC index ▪ Check engine control module system related DTCs and refer to relevant DTC index ▪ Check and install a new engine control module as required ▪ Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine ▪ Using the manufacturer approved diagnostic system carry out 'fuel pump' routine (0x4027_00) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P064F-00	Unauthorized Software/Calibration Detected - No sub type information	▪ Incorrect engine control module calibration for vehicle specification ▪ Corrupt engine control module software download ▪ The engine control module software was modified and the parameters are no longer the designated software parameters	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software on the engine control module ▪ This DTC will never be erased

		designed for this particular engine	
P0652-00	Sensor Reference Voltage "B" Circuit Low - No sub type information	- Other sensor related DTCs 5 volt sensor circuit short circuit to ground, open circuit, high resistance - Turbine intake shut-off valve position sensor - EGR cooler by-pass valve - EGR - Low pressure differential pressure sensor - Fuel rail pressure sensor - Accelerator pedal position sensor - Mass air flow sensor - Camshaft position sensor - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module related DTCs and refer to rele index - Refer to the electrical circuit d and check the 5 volt sensor ci short circuit to ground, open c high resistance - Inspect connectors for signs o ingress, and pins for damage i corrosion - Using the manufacturer appro diagnostic system clear all sto using the Diagnosis Menu tab retest
P0653-00	Sensor Reference Voltage "B" Circuit High - No sub type information	- Other sensor related DTCs 5 volt sensor circuit short circuit to ground, open circuit, high resistance - Turbine intake shut-off valve position sensor - EGR cooler by-pass valve - EGR - Low pressure differential pressure sensor - Fuel rail pressure sensor - Accelerator pedal position sensor - Mass air flow sensor - Camshaft position sensor - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module related DTCs and refer to rele index - Refer to the electrical circuit d and check the 5 volt sensor ci short circuit to ground, open c high resistance - Inspect connectors for signs o ingress, and pins for damage i corrosion - Using the manufacturer appro diagnostic system clear all sto using the Diagnosis Menu tab retest
P0658-00	Actuator Supply Voltage "A" Circuit Low - No sub type	- Vehicle battery under charged - Charging system fault - under	Refer to the workshop manual battery care manual, inspect t battery and ensure it is fully ch

	information	charging - Generator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Excessive quiescent current	and serviceable before performing further tests - Refer to the electrical circuit diagram and check the power and ground connections to the module - Check the vehicle charging system performance to ensure the voltage regulation is correct - Refer to the electrical circuit diagram and check the generator circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0659-00	Actuator Supply Voltage "A" Circuit High - No sub type information	- Other related DTCs - Vehicle battery over charged - Charging system fault - over charging - Generator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module main fuse mechanical integrity - Connector is disconnected, connector pin is backed out, connector pin corrosion - Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance	- Check engine control module related DTCs and refer to relevant index - Check the vehicle charging system performance to ensure the voltage regulation is correct - Refer to the electrical circuit diagram and check the generator circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check engine control module for mechanical integrity - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Refer to the electrical circuit diagram and check connections are secure wiring integrity - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0667-00	Control Module Internal Temperature Sensor	- Other related DTCs - Harness failure - Wiring integrity	Check engine control module related DTCs and refer to relevant index

	"A" Range/Performance - No sub type information	short circuit to ground, short circuit to power, open circuit, high resistance Engine control module internal temperature sensor failure	- Refer to the electrical circuit diagram and check connections are secure and wiring integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the Engine Control Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the Fuel pump' routine (0x4027_00) - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0667-64	Control Module Internal Temperature Sensor "A" Range/Performance - Signal plausibility failure	- The engine control module detected plausibility failures - Other related DTCs - Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module internal temperature sensor failure	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check connections are secure and wiring integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the Engine Control Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the Fuel pump' routine (0x4027_00)
P0667-84	Control Module Internal Temperature Sensor "A" Range/Performance - Signal below allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range - Other related DTCs - Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module internal temperature sensor failure	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check connections are secure and wiring integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out the Engine Control Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out the Fuel pump' routine (0x4027_00)

			pump' routine (0x4027_00
P0668-00	Control Module Internal Temperature Sensor "A" Circuit Low - No sub type information	- Other related DTCs - Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module internal temperature sensor failure	- Check engine control module related DTCs and refer to rele index - Refer to the electrical circuit d and check connections are sec wiring integrity - Check and install a new engin module as required - Using the manufacturer ap diagnostic system carry o Module - Powertrain cont module - Diesel' routine - Using the manufacturer ap diagnostic system carry o pump' routine (0x4027_00
P0668-16	Control Module Internal Temperature Sensor "A" Circuit Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Other related DTCs - Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module internal temperature sensor failure	- Check engine control module related DTCs and refer to rele index - Refer to the electrical circuit d and check connections are sec wiring integrity - Check and install a new engin module as required - Using the manufacturer ap diagnostic system carry o Module - Powertrain cont module - Diesel' routine - Using the manufacturer ap diagnostic system carry o pump' routine (0x4027_00
P0669-00	Control Module Internal Temperature Sensor "A" Circuit High - No sub type information	- Other related DTCs - Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module internal temperature sensor failure	- Check engine control module related DTCs and refer to rele index - Refer to the electrical circuit d and check connections are sec wiring integrity - Check and install a new engin module as required - Using the manufacturer ap diagnostic system carry o Module - Powertrain cont module - Diesel' routine

			Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00)
P0669-17	Control Module Internal Temperature Sensor "A" Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Other related DTCs - Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module internal temperature sensor failure	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check connections are secure and wiring integrity - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system carry out 'Module - Powertrain control module - Diesel' routine - Using the manufacturer approved diagnostic system carry out 'pump' routine (0x4027_00)
P066A-11	Cylinder 1 Glow Plug Control Circuit Low - Circuit short to ground	NOTE: Glow plug control module output pin connected to ground - The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Glow plug circuit short circuit to ground - Connector is disconnected, connector pin is backed out, connector pin corrosion - Glow plug failure	- Refer to the electrical circuit diagram and check the glow plug circuit short circuit to ground - Inspect connectors for signs of ingress, and pins for damage or corrosion - Check and install a new glow plug as required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P066B-12	Cylinder 1 Glow Plug Control Circuit High - Circuit short to battery		- Refer to the electrical circuit diagram and check the glow plug circuit short circuit to power - Inspect connectors for signs of

		NOTE: Glow plug control module output pin connected to power ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Glow plug circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P066C-11	Cylinder 2 Glow Plug Control Circuit Low - Circuit short to ground	NOTE: Glow plug control module output pin connected to ground ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Glow plug circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to ground ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P066D-12	Cylinder 2 Glow Plug Control Circuit High - Circuit short to battery		▪ Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to power ▪ Inspect connectors for signs of

		NOTE: Glow plug control module output pin connected to power ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Glow plug circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P066E-11	Cylinder 3 Glow Plug Control Circuit Low - Circuit short to ground	NOTE: Glow plug control module output pin connected to ground ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Glow plug circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to ground ▪ Inspect connectors for signs of damage, moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P066F-12	Cylinder 3 Glow Plug Control Circuit High - Circuit short to battery		▪ Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to power ▪ Inspect connectors for signs of damage, moisture ingress, and pins for damage or corrosion

		NOTE: Glow plug control module output pin connected to power ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Glow plug circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	ingress, and pins for damage and corrosion ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0670-13	Glow Plug Control Module 1 Control Circuit/Open - Circuit open	NOTE: Glow plug control module ST pin open circuit, high resistance ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Glow plug control module circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagram and check the glow plug control module circuit for open circuit resistance ▪ Inspect connectors for signs of damage, moisture ingress, and pins for damage and corrosion ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0671-13	Cylinder 1 Glow Plug Circuit/Open - Circuit open		▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance

		NOTE: Glow plug control module output pin open circuit, high resistance ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0671-96	Cylinder 1 Glow Plug Circuit/Open - Component internal failure	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module output terminal is open circuit or short circuit to power ▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P0672-13	Cylinder 2 Glow Plug Circuit/Open - Circuit open	NOTE: Glow plug control module output pin open circuit, high resistance ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored fault codes using the Diagnosis Menu tab and retest
P0672-96	Cylinder 2 Glow Plug Circuit/Open - Component internal failure	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module output terminal is open circuit or short circuit to power ▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored fault codes using the Diagnosis Menu tab and retest

P0673-13	Cylinder 3 Glow Plug Circuit/Open - Circuit open	NOTE: Glow plug control module output pin open circuit, high resistance ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0673-96	Cylinder 3 Glow Plug Circuit/Open - Component internal failure	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module output terminal is open circuit or short circuit to power ▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P0674-13	Cylinder 4 Glow Plug Circuit/Open - Circuit open	NOTE: Glow plug control module output pin open circuit, high resistance ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0674-96	Cylinder 4 Glow Plug Circuit/Open - Component internal failure	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module output terminal is open circuit or short circuit to power ▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P0675-13	Cylinder 5 Glow Plug Circuit/Open - Circuit open	NOTE: Glow plug control module output pin open circuit, high resistance ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0675-96	Cylinder 5 Glow Plug Circuit/Open - Component internal failure	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module output terminal is open circuit or short circuit to power ▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P0676-13	Cylinder 6 Glow Plug Circuit/Open - Circuit open	NOTE: Glow plug control module output pin open circuit, high resistance ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0676-96	Cylinder 6 Glow Plug Circuit/Open - Component internal failure	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module output terminal is open circuit or short circuit to power ▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P067A-11	Cylinder 4 Glow Plug Control Circuit Low - Circuit short to ground	NOTE: Glow plug control module output pin connected to ground - The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Glow plug circuit short circuit to ground - Connector is disconnected, connector pin is backed out, connector pin corrosion - Glow plug failure	- Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to ground - Inspect connectors for signs of damage, ingress, and pins for damage or corrosion - Check and install a new glow plug if required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P067B-12	Cylinder 4 Glow Plug Control Circuit High - Circuit short to battery	NOTE: Glow plug control module output pin connected to power - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Glow plug circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Glow plug failure	- Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to power - Inspect connectors for signs of damage, ingress, and pins for damage or corrosion - Check and install a new glow plug if required - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P067C-11	Cylinder 5 Glow Plug Control Circuit Low - Circuit short to ground		- Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to ground - Inspect connectors for signs of damage, ingress, and pins for damage or corrosion

		NOTE: Glow plug control module output pin connected to ground ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Glow plug circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P067D-12	Cylinder 5 Glow Plug Control Circuit High - Circuit short to battery	NOTE: Glow plug control module output pin connected to power ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Glow plug circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to power ▪ Inspect connectors for signs of damage, moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P067E-11	Cylinder 6 Glow Plug Control Circuit Low - Circuit short to ground		▪ Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to ground ▪ Inspect connectors for signs of damage, moisture ingress, and pins for damage or corrosion

		NOTE: Glow plug control module output pin connected to ground ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Glow plug circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	ingress, and pins for damage and corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P067F-11	Cylinder 6 Glow Plug Control Circuit High - Circuit short to ground	NOTE: Glow plug control module output pin connected to power ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Glow plug circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to power ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Check and install a new glow plug if required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P067F-12	Cylinder 6 Glow Plug Control Circuit High - Circuit short to battery		▪ Refer to the electrical circuit diagram and check the glow plug circuit for short circuit to power ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion

		NOTE: Glow plug control module output pin connected to power ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Glow plug circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure	corrosion ▪ Check and install a new glow required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0683-88	Glow Plug Control Module 1 to PCM Communication Circuit - Bus off	NOTE: Glow plug control module Diag pin open circuit, high resistance ▪ The engine control module has determined failures where a data bus is not available ▪ Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagram and check the glow plug control module circuit for short circuit to ground, short circuit to power, short circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage or corrosion ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0686-11	ECM/PCM Power Relay Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when	▪ Refer to the electrical circuit diagram and check engine management relay power and ground circuit for open circuit ▪ Refer to the electrical circuit diagram

		another value was expected ▪ Engine management system relay power and ground circuits open circuit ▪ Engine management system relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine management system relay failure	and check the engine management system relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Check and install a new engine management system relay as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0687-73	ECM/PCM Power Relay Control Circuit High - Actuator stuck closed	▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Engine management system relay power and ground circuits open circuit ▪ Engine management system relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine management system relay failure	▪ Refer to the electrical circuit diagram and check engine management system relay power and ground circuit for open circuit ▪ Refer to the electrical circuit diagram and check the engine management system relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Check and install a new engine management system relay as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P068A-00	ECM/PCM Power Relay De-Energized - Too Early - No sub type information	▪ Vehicle battery disconnected before engine management system relay has powered down ▪ Engine management system relay power and ground circuits open circuit ▪ Engine management system relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ This DTC is set when the engine management system high current contacts open early - Indicating a hold failure. Check the vehicle battery has not been disconnected before the engine management system relay powered down. Check the operation of the engine management system current relay ▪ Refer to the electrical circuit diagram and check engine management system relay power and ground circuit for open circuit

		Engine management system relay failure	- Refer to the electrical circuit diagram and check the engine management system relay circuit for short circuit to ground, short circuit to power circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Check and install a new engine management system relay as required - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0691-11	Fan 1 Control Circuit Low - Circuit short to ground	- The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Main electric/viscous cooling fan circuit short circuit to ground, open circuit, high resistance - Main electric/viscous cooling fan ignition powerfeed circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Main electric/viscous cooling fan failure	- Refer to the electrical circuit diagram and check the main electric/viscous cooling fan circuit for short circuit to ground, open circuit, high resistance - Refer to the electrical circuit diagram and check the main electric/viscous cooling fan ignition powerfeed circuit for short open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Check and install a new main electric/viscous cooling fan as required - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P0691-16	Fan 1 Control Circuit Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Main electric cooling fan circuit short circuit to ground, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Main electric cooling fan failure	- Refer to the electrical circuit diagram and check the main electric cooling fan circuit for short circuit to ground, open circuit, high resistance - Inspect connectors for signs of moisture ingress, and pins for damage and corrosion - Check and install a new main electric cooling fan as required - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest

P0692-12	Fan 1 Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Main electric/viscous cooling fan circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Main electric/viscous cooling fan failure	▪ Refer to the electrical circuit diagram and check the main electric/viscous cooling fan circuit for short circuit to power ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new main electric/viscous cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0692-17	Fan 1 Control Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Main electric cooling fan circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Main electric cooling fan failure	▪ Refer to the electrical circuit diagram and check the main electric cooling fan circuit for short circuit to power ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new main electric cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0693-11	Fan 2 Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Auxiliary electric cooling fan circuit short circuit to ground, open circuit, high resistance ▪ Auxiliary electric cooling fan ignition powerfeed circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Auxiliary electric cooling fan failure	▪ Refer to the electrical circuit diagram and check the auxiliary electric cooling fan circuit for short circuit to ground, open circuit, high resistance ▪ Refer to the electrical circuit diagram and check the auxiliary electric cooling fan ignition powerfeed circuit for open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new auxiliary electric cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

P0693-16	Fan 2 Control Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Auxiliary electric cooling fan circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Auxiliary electric cooling fan failure	▪ Refer to the electrical circuit diagram and check the auxiliary electric fan circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Check and install a new auxiliary electric cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0694-12	Fan 2 Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Auxiliary electric cooling fan circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Auxiliary electric cooling fan failure	▪ Refer to the electrical circuit diagram and check the auxiliary electric fan circuit for short circuit to power ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Check and install a new auxiliary electric cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0694-17	Fan 2 Control Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Auxiliary electric cooling fan circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Auxiliary electric cooling fan failure	▪ Refer to the electrical circuit diagram and check the auxiliary electric fan circuit for short circuit to power ▪ Inspect connectors for signs of moisture ingress, and pins for damage and corrosion ▪ Check and install a new auxiliary electric cooling fan as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P0698-00	Sensor Reference Voltage "C" Circuit	▪ Other sensor related DTCs	▪ Check engine control module related DTCs and refer to relevant information

	Low - No sub type information	5 volt sensor circuit short circuit to ground, open circuit, high resistance - Compressor recirculation valve - Differential pressure sensor - EGR - Low pressure butterfly valve - Crankcase ventilation pressure sensor - Connector is disconnected, connector pin is backed out, connector pin corrosion	index - Refer to the electrical circuit diagram and check the 5 volt sensor circuit for short circuit to ground, open circuit or high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P0699-00	Sensor Reference Voltage "C" Circuit High - No sub type information	- Other sensor related DTCs 5 volt sensor circuit short circuit to ground, open circuit, high resistance - Compressor recirculation valve - Differential pressure sensor - EGR - Low pressure butterfly valve - Crankcase ventilation pressure sensor - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check the 5 volt sensor circuit for short circuit to ground, open circuit or high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06A6-1C	Sensor Reference Voltage "A" Circuit Range/Performance - Circuit voltage out of range	- The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low - Other sensor related DTCs 5 volt sensor circuit short circuit to ground, open circuit, high resistance - EGR valve - High pressure - EGR valve - Low pressure - Manifold absolute pressure and temperature sensor - Electric throttle - Charge air pressure sensor	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check the 5 volt sensor circuit for short circuit to ground, open circuit or high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest

		- Variable geometry turbocharger - Primary - Accelerator pedal position sensor - Crankshaft position sensor - Connector is disconnected, connector pin is backed out, connector pin corrosion	
P06A7-1C	Sensor Reference Voltage "B" Circuit Range/Performance - Circuit voltage out of range	- The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low - Other sensor related DTCs 5 volt sensor circuit short circuit to ground, open circuit, high resistance - Turbine intake shut-off valve position sensor - EGR cooler by-pass valve - EGR - Low pressure differential pressure sensor - Fuel rail pressure sensor - Accelerator pedal position sensor - Mass air flow sensor - Camshaft position sensor - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check the 5 volt sensor circuit for short circuit to ground, open circuit or high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system clear all stored codes and retest
P06A8-1C	Sensor Reference Voltage "C" Circuit Range/Performance - Circuit voltage out of range	- The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low - Other sensor related DTCs 5 volt sensor circuit short circuit to ground, open circuit, high resistance - Compressor recirculation valve - Differential pressure sensor - EGR - Low pressure butterfly	- Check engine control module related DTCs and refer to relevant index - Refer to the electrical circuit diagram and check the 5 volt sensor circuit for short circuit to ground, open circuit or high resistance - Inspect connectors for signs of moisture ingress, and pins for damage or corrosion - Using the manufacturer approved diagnostic system clear all stored codes and retest

		valve ▪ Crankcase ventilation pressure sensor ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	using the Diagnosis Menu tab retest
P06B3-64	Sensor Power Supply "B" Circuit/Open - Signal plausibility failure	▪ The engine control module detected plausibility failures ▪ The engine control module has not received the expected CAN signal from the gateway module within the specified time interval ▪ Other gateway module related DTCs ▪ CAN harness link between engine control module and gateway module network malfunction ▪ Gateway module power and ground circuits open circuit	▪ Check gateway module for related DTCs and refer to relevant DTCs ▪ Using the manufacturer approved diagnostic system, perform a network integrity test ▪ Refer to the electrical circuit diagram and check gateway module power and ground circuits for open circuit ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P06B3-82	Sensor Power Supply "B" Circuit/Open - Alive / sequence counter incorrect / not updated	▪ The engine control module has indicated that a signal was received without the corresponding rolling count value being properly updated ▪ The engine control module has not received the expected CAN signal from the gateway module within the specified time interval ▪ Other gateway module related DTCs ▪ CAN harness link between engine control module and gateway module network malfunction ▪ Gateway module power and ground circuits open circuit	▪ Check gateway module for related DTCs and refer to relevant DTCs ▪ Using the manufacturer approved diagnostic system, perform a network integrity test ▪ Refer to the electrical circuit diagram and check gateway module power and ground circuits for open circuit ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest
P06B9-1E	Cylinder 1 Glow Plug Circuit Range/Performance - Circuit resistance out of range		▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of corrosion, ingress, and pins for damage

		NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module internal logic ■ The engine control module has measured a resistance outside the expected range but not identified as too high or too low ■ Glow plug circuit open circuit, high resistance ■ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ■ Glow plug failure ■ Glow plug control module failure	corrosion ■ Check and install a new glow required ■ Check and install a new glow control module as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06BA-1E	Cylinder 2 Glow Plug Circuit Range/Performance - Circuit resistance out of range	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module internal logic ■ The engine control module has measured a resistance outside the expected range but not identified as too high or too low ■ Glow plug circuit open circuit, high resistance ■ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ■ Glow plug failure ■ Glow plug control module failure	■ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ■ Inspect connectors for signs of damage, ingress, and pins for damage or corrosion ■ Check and install a new glow plug required ■ Check and install a new glow plug control module as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06BB-1E	Cylinder 3 Glow Plug Circuit		■ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance

	Range/Performance - Circuit resistance out of range	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module internal logic ▪ The engine control module has measured a resistance outside the expected range but not identified as too high or too low ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure ▪ Glow plug control module failure	open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new glow plug control module as required ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06BC-1E	Cylinder 4 Glow Plug Circuit Range/Performance - Circuit resistance out of range	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module internal logic ▪ The engine control module has measured a resistance outside the expected range but not identified as too high or too low ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new glow plug control module as required ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06BD-1E	Cylinder 5 Glow Plug Circuit		▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance

	Range/Performance - Circuit resistance out of range	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module internal logic ▪ The engine control module has measured a resistance outside the expected range but not identified as too high or too low ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure ▪ Glow plug control module failure	open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new glow plug control module as required ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06BE-1E	Cylinder 6 Glow Plug Circuit Range/Performance - Circuit resistance out of range	NOTE: To monitor glow plug and glow plug wiring harness. Glow plug control module internal logic ▪ The engine control module has measured a resistance outside the expected range but not identified as too high or too low ▪ Glow plug circuit open circuit, high resistance ▪ Glow plug control module connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Glow plug failure ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagram and check the glow plug circuit for open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new glow plug control module as required ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06DA-13	Engine Oil Pressure Control	▪ The engine control module has determined an open circuit via	▪ Refer to the electrical circuit diagram and check the electric oil pump

	Circuit/Open - Circuit open	lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Electric oil pump solenoid circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Electric oil pump solenoid failure	solenoid circuit for open circuit resistance ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new electric oil pump solenoid as required ▪ Only if an oil change occurs or after repair. Using the manufacturer approved diagnostic system carry out 'Oil service counter reset' routine (0x4) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P06DB-11	Engine Oil Pressure Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Electric oil pump solenoid circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Electric oil pump solenoid failure	▪ Refer to the electrical circuit diagram and check the electric oil pump solenoid circuit for short circuit to ground ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new electric oil pump solenoid as required ▪ Only if an oil change occurs or after repair. Using the manufacturer approved diagnostic system carry out 'Oil service counter reset' routine (0x4) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P06DC-12	Engine Oil Pressure Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Electric oil pump solenoid circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out,	▪ Refer to the electrical circuit diagram and check the electric oil pump solenoid circuit for short circuit to power ▪ Inspect connectors for signs of ingress, and pins for damage or corrosion ▪ Check and install a new electric oil pump solenoid as required ▪ Only if an oil change occurs or after repair. Using the

		connector pin corrosion ▪ Electric oil pump solenoid failure	manufacturer approved diagnostic system carry out 'Oil service counter reset' routine (0x4) ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P06DF-96	Glow Plug Control Module 1 Memory Checksum Error - Component internal failure	▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ▪ LIN_B circuit and Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ LIN_B circuit or Glow plug control module failure	▪ Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new glow plug control module as required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P06EA-00	NOx Sensor Processor Performance Bank 1 Sensor 1 - No sub type information	▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	▪ Refer to the electrical circuit diagram and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of ingress, and pins for damage and corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P06EB-00	NOx Sensor Processor Performance Bank 1 Sensor 2 - No sub type information	▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected,	▪ Refer to the electrical circuit diagram and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		connector pin is backed out, connector pin corrosion Post selective catalytic reduction NOx sensor failure	- Inspect connectors for signs of ingress, and pins for damage and corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P06F5-16	Glow Plug Control Module 2 - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - LIN_B circuit and Glow plug control module circuit short circuit to ground, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - LIN_B circuit or Glow plug control module failure	- Refer to the electrical circuit diagram and check the LIN_B circuit and glow plug control module circuit for circuit to ground, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check and install a new glow plug control module as required - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P06F5-17	Glow Plug Control Module 2 - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - LIN_B circuit and Glow plug control module circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - LIN_B circuit or Glow plug control module failure	- Refer to the electrical circuit diagram and check the LIN_B circuit and glow plug control module circuit for circuit to power - Inspect connectors for signs of ingress, and pins for damage and corrosion - Check and install a new glow plug control module as required - Using the manufacturer approved diagnostic system clear all stored faults using the Diagnosis Menu tab and retest
P06F5-96	Glow Plug Control Module 2 - Component internal failure	The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault	- Refer to the electrical circuit diagram and check the LIN_B circuit and glow plug control module circuit for circuit to ground, short circuit, open circuit, high resistance - Inspect connectors for signs of ingress, and pins for damage and corrosion

		■ LIN_B circuit and Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ LIN_B circuit or Glow plug control module failure	ingress, and pins for damage corrosion ■ Check and install a new glow control module as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06F9-13	Glow Plug Control Module 2 System Voltage - Circuit open	■ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ■ LIN_B circuit and Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ LIN_B circuit or Glow plug control module failure	■ Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of damage, ingress, and pins for damage corrosion ■ Check and install a new glow control module as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06F9-16	Glow Plug Control Module 2 System Voltage - Circuit voltage below threshold	■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ LIN_B circuit and Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ LIN_B circuit or Glow plug control module failure	■ Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of damage, ingress, and pins for damage corrosion ■ Check and install a new glow control module as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06F9-17	Glow Plug Control Module 2 System	■ The engine control module measured a voltage above a	■ Refer to the electrical circuit diagram and check the LIN_B circuit or

	Voltage - Circuit voltage above threshold	specified range but not necessarily a short circuit to power ■ LIN_B circuit and Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ LIN_B circuit or Glow plug control module failure	plug control module circuit for circuit to ground, short circuit open circuit, high resistance ■ Inspect connectors for signs of ingress, and pins for damage or corrosion ■ Check and install a new glow plug control module as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest
P06FB-96	Glow Plug Control Module 2 Memory Checksum Error - Component internal failure	■ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ■ LIN_B circuit and Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ LIN_B circuit or Glow plug control module failure	■ Refer to the electrical circuit diagram and check the LIN_B circuit or glow plug control module circuit for circuit to ground, short circuit open circuit, high resistance ■ Inspect connectors for signs of ingress, and pins for damage or corrosion ■ Check and install a new glow plug control module as required ■ Using the manufacturer approved diagnostic system clear all stored codes using the Diagnosis Menu tab and retest